Taiwan Semiconductor Manufacturing Co.

Conviction grows as AI demand expands; advanced capacity races to catch up

Reiterate Rating: BUY | PO: 3,260 TWD | Price: 2,440 TWD

Resilient GM outlook amid 2nm ramp & depreciation hike

TSMC reported strong 3Q26 results, guided 4Q26 sales at +11-16% QoQ, lifting full year sales to 44% YoY in US$ on steep 2nm ramp for Apple and AMD and even higher 3nm utilization from GPUs/ASICs refreshes and ARM-based CPU demand uptick. Despite dilution from further 2nm ramp and +28% HoH depreciation in 2H26 balancing favorable utilization and pricing, GMs is guided to hold stable QoQ at 66% +/-1% in 4Q26.

Accelerating expansion to support proliferating drivers

Reflecting TSMC's expectation of stronger AI accelerator chipset demand (vs. its guide of 55-59% CAGR) supplemented by the agentic AI growth (e.g. CPUs and networking switches), our demand driver bottom-up model shows its sales could grow 35% CAGR from '24-28 (vs. its guide of ~25% CAGR). To enable the growth and narrow the supply and demand gap, TSMC is pulling in the equipment and facility production schedule. Factoring in an aggressive ramp, we model its capex will need to step up to US$83bn/US$90bn in 2027/28 to support a ~50% CAGR for its 3nm and below nodes capacity growth from '25-28 and 2.5D/3D packaging expansion. Factoring utilization and pricing benefit partially offset by +24% CAGR depreciation, new node ramp/overseas fab dilution and cost inflation, its GMs could still expand modestly to 67-68% by '28.

Robust outlook reaffirming AI and capex sustainability

Factoring in stronger business outlook on strong technology and scale leadership, extending AI demand visibility and firmer advanced/mature pricing outlook more than offsetting the near term margins dip, we tweak our '26/27/28 EPS to NT$112/159/201 (vs. NT$112/161/203 prior). We maintain our Buy rating with PO unchanged at NT$3,260/US$620 on 20x '27 P/E. The share valuation is undemanding at 15x '27 P/E (vs. its long term 11-21x range) backed by 31%/34% sales/earnings CAGR in '26-28.

Estimates (Dec) (NT$)	2024A	2025A	2026E	2027E	2028E
Net Income (Adjusted - mn)	1,173,268	1,717,886	2,901,479	4,125,623	5,212,314
EPS	45.24	66.24	111.88	159.09	200.99
EPS Change (YoY)	39.9%	46.4%	68.9%	42.2%	26.3%
Consensus EPS (Visible Alpha)			111.80	155.30	205.37
Dividend / Share	18.00	21.50	24.00	24.00	24.00
Free Cash Flow / Share	33.55	38.66	65.84	94.43	136.16
ADR EPS (US$)	7.04	10.62	17.42	24.77	31.30
ADR Dividend / Share (US$)	2.80	3.45	3.74	3.74	3.74
Valuation (Dec)					
P/E	53.93x	36.83x	21.81x	15.34x	12.14x
Dividend Yield	0.738%	0.881%	0.984%	0.984%	0.984%
EV / EBITDA*	30.30x	22.91x	14.59x	10.19x	8.27x
Free Cash Flow Yield*	1.38%	1.58%	2.70%	3.87%	5.58%

* For full definitions of iQmethod℠ measures, see page 21.

This research report provides general information only. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or other persons without the express written consent of BofA Securities.
>> Employed by a non-US affiliate of BofAS and is not registered/qualified as a research analyst under the FINRA rules.
Refer to "Other Important Disclosures" for information on certain BofA Securities entities that take responsibility for the information herein in particular jurisdictions.
BofA Securities does and seeks to do business with issuers covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.
Refer to important disclosures on page 22 to 25. Analyst Certification on page 19. Price Objective Basis/Risk on page 19.
13028739

Timestamp: 15 October 2026 01:52PM EDT

16 October 2026

Equity

Key Changes

(NT$)	Previous	Current
2026E EPS	112.46	111.88
2027E EPS	161.36	159.09
2028E EPS	203.48	200.99
2026E EBITDA (m)	4,155,258.3	4,122,659.2
2027E EBITDA (m)	5,927,984.9	5,899,564.6
2028E EBITDA (m)	7,307,336.4	7,272,267.8

Haas Liu >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3727
haas.liu@bofa.com

Mike Yang >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3729
mike.c.yang@bofa.com

Robert Cheng >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3731
robert.cheng@bofa.com

Cathy Hsu >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3726
cathy.hsu3@bofa.com

Stock Data

Price (Common / ADR)	2,440 TWD / 430.00 USD
Price Objective	3,260 TWD / 620.00 USD
Date Established	12-Oct-2026 / 12-Oct-2026
Investment Opinion	B-1-7 / B-1-7
52-Week Range	1,375 TWD-2,590 TWD
Market Value (mn)	1,970,784 USD
Market Value (mn)	63,274,983 TWD
Shares Outstanding (mn)	25,932.4 / 5,186.5
Average Daily Value (mn)	1,780 USD
Free Float	92.2%
BofA Ticker / Exchange	TSMWF / TAI
BofA Ticker / Exchange	TSM / NYS
Bloomberg / Reuters	2330 TT / 2330.TW
ROE (2026E)	44.0%
Net Dbt to Eqty (Dec-2025A)	-37.3%

Glossary page 18

iQprofile℠ Taiwan Semiconductor Manufacturing Co.

Key Income Statement Data (Dec)	2024A	2025A	2026E	2027E	2028E
(NT$ Millions)					
Sales	2,894,308	3,809,054	5,594,068	7,852,781	9,653,190
Gross Profit	1,624,354	2,281,294	3,728,119	5,331,822	6,583,408
Sell General & Admin Expense	(96,888)	(99,222)	(109,745)	(126,674)	(137,436)
Operating Profit	1,322,053	1,936,092	3,290,950	4,777,816	6,000,060
Net Interest & Other Income	83,786	105,574	195,240	173,467	257,959
Associates	NA	NA	NA	NA	NA
Pretax Income	1,405,839	2,041,666	3,486,190	4,951,283	6,258,020
Tax (expense) / Benefit	(233,407)	(326,266)	(584,506)	(828,600)	(1,049,420)
Net Income (Adjusted)	1,173,268	1,717,886	2,901,479	4,125,623	5,212,314
Average Fully Diluted Shares Outstanding	25,932	25,933	25,933	25,933	25,933
Key Cash Flow Statement Data					
Net Income	1,173,268	1,717,886	2,901,479	4,125,623	5,212,314
Depreciation & Amortization	662,797	688,096	831,710	1,121,749	1,272,207
Change in Working Capital	(143,010)	(72,100)	(111,287)	(169,227)	(104,410)
Deferred Taxation Charge	NA	NA	NA	NA	NA
Other Adjustments, Net	133,123	(58,907)	73,153	28,040	25,528
Cash Flow from Operations	1,826,177	2,274,976	3,695,054	5,106,185	6,405,639
Capital Expenditure	(956,007)	(1,272,411)	(1,987,705)	(2,657,265)	(2,874,767)
(Acquisition) / Disposal of Investments	(58,000)	(37,014)	(57,395)	3,030	189
Other Cash Inflow / (Outflow)	149,164	165,031	203,847	2,716	2,948
Cash Flow from Investing	(864,843)	(1,144,393)	(1,841,253)	(2,651,518)	(2,871,629)
Shares Issue / (Repurchase)	(9)	(2)	(1)	0	0
Cost of Dividends Paid	(337,129)	(440,855)	(531,614)	(622,377)	(622,377)
Cash Flow from Financing	(346,301)	(440,345)	(610,136)	(624,653)	(623,069)
Free Cash Flow	870,171	1,002,565	1,707,349	2,448,920	3,530,872
Net Debt	(1,403,733)	(2,035,607)	(3,391,133)	(5,225,188)	(8,136,931)
Change in Net Debt	(556,872)	(647,079)	(1,239,614)	(1,832,290)	(2,911,633)
Key Balance Sheet Data					
Property, Plant & Equipment	3,234,980	3,691,841	5,010,432	6,574,058	8,034,129
Other Non-Current Assets	368,606	424,052	474,902	469,917	469,137
Trade Receivables	272,088	281,791	343,813	433,589	481,703
Cash & Equivalents	2,422,020	3,068,595	4,434,206	6,265,984	9,177,035
Other Current Assets	394,245	466,745	536,648	653,596	746,634
Total Assets	6,691,938	7,933,024	10,800,000	14,397,145	18,908,638
Long-Term Debt	958,429	896,062	874,628	863,062	860,183
Other Non-Current Liabilities	145,408	118,147	205,715	254,787	291,926
Short-Term Debt	59,858	136,926	168,444	177,735	179,921
Other Current Liabilities	1,204,667	1,321,094	1,736,885	1,790,702	1,843,926
Total Liabilities	2,368,362	2,472,229	2,985,672	3,086,286	3,175,956
Total Equity	4,323,576	5,460,795	7,814,328	11,310,859	15,732,682
Total Equity & Liabilities	6,691,938	7,933,024	10,800,000	14,397,145	18,908,638
iQ*method℠ - Bus Performance					
Return On Capital Employed	23.2%	28.4%	36.3%	38.1%	35.1%
Return On Equity	30.3%	35.4%	44.0%	43.3%	38.7%
Operating Margin	45.7%	50.8%	58.8%	60.8%	62.2%
EBITDA Margin	68.6%	68.9%	73.7%	75.1%	75.3%
iQ*method℠ - Quality of Earnings					
Cash Realization Ratio	1.6x	1.3x	1.3x	1.2x	1.2x
Asset Replacement Ratio	1.4x	1.8x	2.4x	2.4x	2.3x
Tax Rate (Reported)	16.6%	16.0%	16.8%	16.7%	16.8%
Net Debt-to-Equity Ratio	-32.5%	-37.3%	-43.4%	-46.2%	-51.7%
Interest Cover	NM	NM	NM	NM	NM

Key Metrics

* For full definitions of *iQmethod*℠ measures, see page 21.

Company Sector

Semiconductors

Company Description

TSMC was founded in 1987 and is listed in Taiwan (2330TT) and the US (TSM/NYS).TSMC is the largest and global leader in integrated circuit (IC) manufacturing. As a build-to-order foundry, it provides a wide range of value-add activities: IC manufacturing, mask-making, IC design services, turnkey solutions, and process development. We attribute its success to its proven, winning business model, unparalleled scale advantage, optimized execution, and technology scope and depth. 5 shares = 1 ADR.

Investment Rationale

The ongoing semi content growth in mobile, rise of artificial intelligence (AI), and proliferation of Internet of Things (IoT) should result in sustainable upside in aggregate computing power globally. TSMC, as the leading contract manufacturer of semiconductor chips, is in a good position to capitalize. We are structurally positive on TSMC and expect it to sustain over 15% growth with rising FCF from 2026 onwards.

Stock Data

Shares / ADR	5.00
Price to Book Value	8.1x

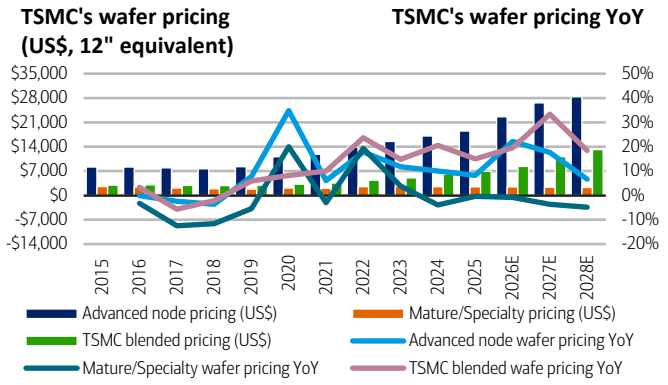
Key Changes

(US$)	Previous	Current
2026E EPS	17.51	17.42
2027E EPS	25.13	24.77
2028E EPS	31.69	31.30

CR

Focus charts and tables

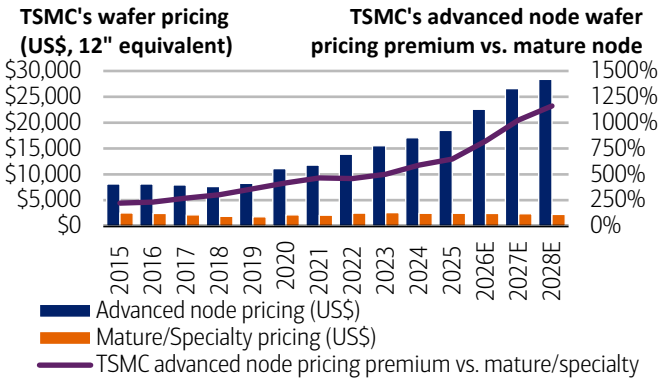
Exhibit 1: TSMC's wafer pricing by advanced vs. mature nodes
TSMC's wafer pricing by advanced vs. mature nodes



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

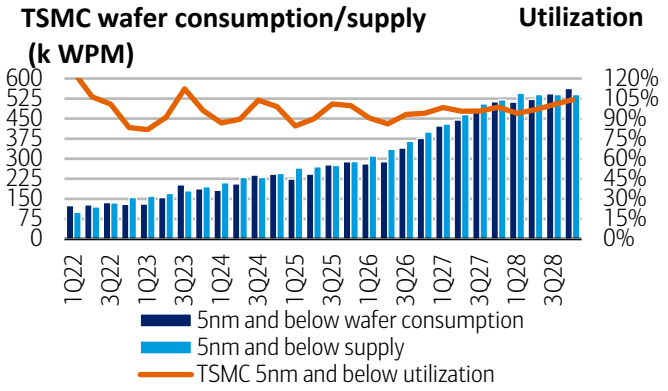
Exhibit 2: TSMC's wafer pricing gap for advanced vs. mature nodes
TSMC's advanced node pricing is now 9x higher than mature ones



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

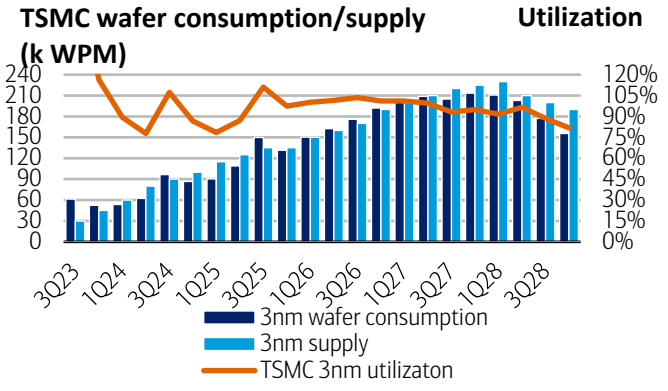
Exhibit 3: TSMC 5nm and below wafer consumption/supply/utilization
Utilization of 5nm and blow to remain high



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

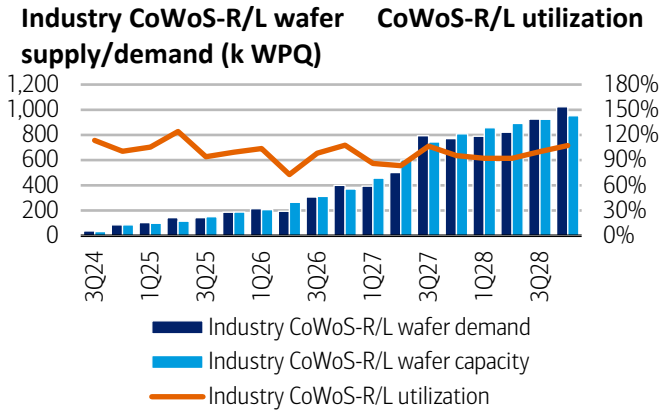
Exhibit 4: TSMC 3nm wafer consumption/supply/utilization
TSMC 3nm to see utilization staying close to full



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

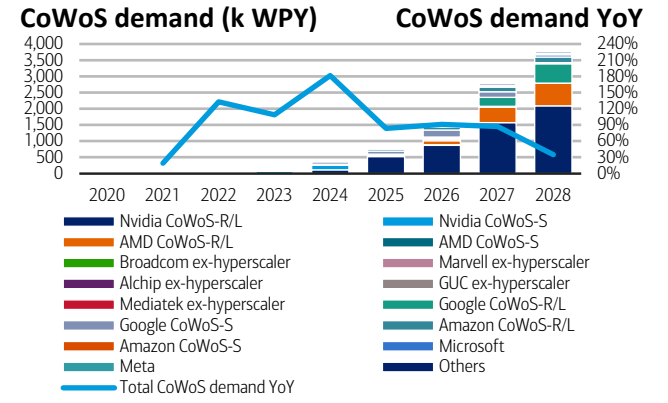
Exhibit 5: Industry CoWoS-R/L supply/demand
CoWoS-R/L utilization should stay at high levels



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 6: Industry CoWoS demand
CoWoS demand should remain strong through 2028



Source: Global Research estimates

BofA GLOBAL RESEARCH

CR

HPC/Apple strength lifts 4Q26 sales; GMs hold stable as 2nm ramps further

3Q26 sales of +16% QoQ is above its US$ guide led by HPC (+22%, across GPUs, CPUs, ASICs, networking switches), smartphone (+10%, Apple seasonal builds) and incremental auto strength (+23%, US and China EV/ADAS ramp). Improving cost and better utilization is partially offset by 2nm ramp, taking 3Q26 GMs down by 130bps QoQ to 66.4% in line with its guidance and BofA preview but below recent street 67-68%. With higher revenue scale partially offset by mild decline in GMs/OpMs, 3Q26 EPS came in at NT$29.43 (vs. BofAe/street NT$29.67/NT$28.38).

Exhibit 7: TSMC 3Q26-4Q26 and 2026-27 estimates BofAe vs. street
We now forecast EPS for '26/27 to be NT$112/159

	3Q26E			4Q26E			2026E			2027E		
	Actual	Old	Street	BofAe	Old	Street	BofAe	Old	Street	BofAe	Old	Street
Sales (NT$mn)	$1,494,243	$1,494,243	$1,463,320	$1,695,341	$1,652,537	$1,607,454	$5,594,068	$5,551,264	$5,458,697	$7,852,781	$7,790,371	$7,429,732
Sales QoQ / YoY (%)	17.6%	17.6%	15.2%	13.5%	10.6%	9.8%	46.9%	45.7%	43.3%	40.4%	40.3%	36.1%
Gross profit (NT$mn)	$992,379	$1,000,465	$972,771	$1,124,134	$1,120,993	$1,062,013	$3,728,119	$3,733,064	$3,626,431	$5,331,822	$5,371,382	$4,942,035
GMs (%)	66.4%	67.0%	66.5%	66.3%	67.8%	66.1%	66.6%	67.2%	66.4%	67.9%	68.9%	66.5%
Operating profit (NT$mn)	$876,445	$891,981	$854,354	$988,936	$1,005,999	$932,497	$3,290,950	$3,323,549	$3,200,560	$4,777,816	$4,866,929	$4,370,091
OpMs (%)	58.7%	59.7%	58.4%	58.3%	60.9%	58.0%	58.8%	59.9%	58.6%	60.8%	62.5%	58.8%
Pretax profit (NT$mn)	$911,039	$924,044	$881,845	$1,024,921	$1,042,707	$964,125	$3,486,190	$3,516,980	$3,380,594	$4,951,283	$5,042,211	$4,529,536
Pretax margins (%)	61.0%	61.8%	60.3%	60.5%	63.1%	60.0%	62.3%	63.4%	61.9%	63.1%	64.7%	61.0%
Tax expense (NT$mn)	-$147,975	-$154,691	-$143,280	-$165,548	-$174,556	-$159,994	-$584,711	-$600,435	-$583,761	-$825,661	-$857,545	-$772,586
Tax rate (%)	16.2%	16.7%	16.2%	16.2%	16.7%	16.6%	16.8%	17.1%	17.3%	16.7%	17.0%	17.1%
Net profit (NT$mn)	$763,064	$769,353	$738,565	$859,373	$868,150	$804,131	$2,901,479	$2,916,545	$2,796,833	$4,125,623	$4,184,666	$3,756,950
Net margins (%)	51.1%	51.5%	50.5%	50.7%	52.5%	50.0%	51.9%	52.5%	51.2%	52.5%	53.7%	50.6%
EPS (NT$)	$29.43	$29.67	$28.38	$33.14	$33.48	$31.13	$111.89	$112.47	$108.02	$159.09	$161.37	$141.20

Source: BofA Global Research estimates, Bloomberg

BofA GLOBAL RESEARCH

For 4Q26, we model TSMC's sales to +13.5% QoQ (vs. its guidance of +11-16% QoQ in NT$, BofAe was +11% QoQ) on further HPC strength and continued flagship smartphone builds mainly for Apple. We expect a mild growth in its wafer shipment (BofAe +13%/-5% QoQ for advanced/mature nodes), balancing HPC demand strength and Apple's chipset builds for its new product launches and muted non-AI related demand.

The blended ASPs strength (BofAe +13% QoQ) should be the key sales driver for 4Q26 with shipment mix continuing shifting toward 3nm and 2nm which in combination could represent ~50% of its 4Q26 sales based on our estimate. Factoring higher depreciation and a 3-4% GMs dilution from further 2nm ramp, we model 4Q26 GM to stay stable QoQ at 66.3% (vs. its 65-67% guide and BofAe prior/recent street 67.8%/68-70%).

Exhibit 8: TSMC 2026-28 estimates BofAe vs. street
We expect TSMC's 2027/2028 sales to grow at 40.4%/22.9% YoY

	2026E				2027E				2028E			
	BofAe	Old	Street	vs. street	BofAe	Old	Street	vs. street	BofAe	Old	Street	vs. street
Sales (NT$mn)	$5,594,068	$5,551,264	$5,458,697	2%	$7,852,781	$7,790,371	$7,429,732	6%	$9,653,190	$9,656,518	$9,403,956	3%
Sales QoQ / YoY (%)	46.9%	45.7%	43.3%		40.4%	40.3%	36.1%		22.9%	24.0%	26.6%	
Gross profit (NT$mn)	$3,728,119	$3,733,064	$3,626,431	3%	$5,331,822	$5,371,382	$4,942,035	8%	$6,583,408	$6,653,851	$6,261,248	5%
GMs (%)	66.6%	67.2%	66.4%	0%	67.9%	68.9%	66.5%	1%	68.2%	68.9%	66.6%	2%
Operating profit (NT$mn)	$3,290,950	$3,323,549	$3,200,560	3%	$4,777,816	$4,866,929	$4,370,091	9%	$6,000,060	$6,097,850	$5,570,732	8%
OpMs (%)	58.8%	59.9%	58.6%	0%	60.8%	62.5%	58.8%	2%	62.2%	63.1%	59.2%	3%
Pretax profit (NT$mn)	$3,486,190	$3,516,980	$3,380,594	3%	$4,951,283	$5,042,211	$4,529,536	9%	$6,258,020	$6,359,178	$5,767,917	8%
Pretax margins (%)	62.3%	63.4%	61.9%	0%	63.1%	64.7%	61.0%	2%	64.8%	65.9%	61.3%	3%
Tax expense (NT$mn)	-$584,711	-$600,435	-$583,761	0%	-$825,661	-$857,545	-$772,586	7%	-$1,045,706	-$1,082,440	-$970,240	8%
Tax rate (%)	16.8%	17.1%	17.3%	0%	16.7%	17.0%	17.1%	0%	16.7%	17.0%	16.8%	0%
Net profit (NT$mn)	$2,901,479	$2,916,545	$2,796,833	4%	$4,125,623	$4,184,666	$3,756,950	10%	$5,212,314	$5,276,738	$4,797,677	9%
Net margins (%)	51.9%	52.5%	51.2%	1%	52.5%	53.7%	50.6%	2%	54.0%	54.6%	51.0%	3%
EPS (NT$)	$111.89	$112.47	$108.02	4%	$159.09	$161.37	$141.20	13%	$201.00	$203.48	$179.44	12%

Source: BofA Global Research estimates, Bloomberg

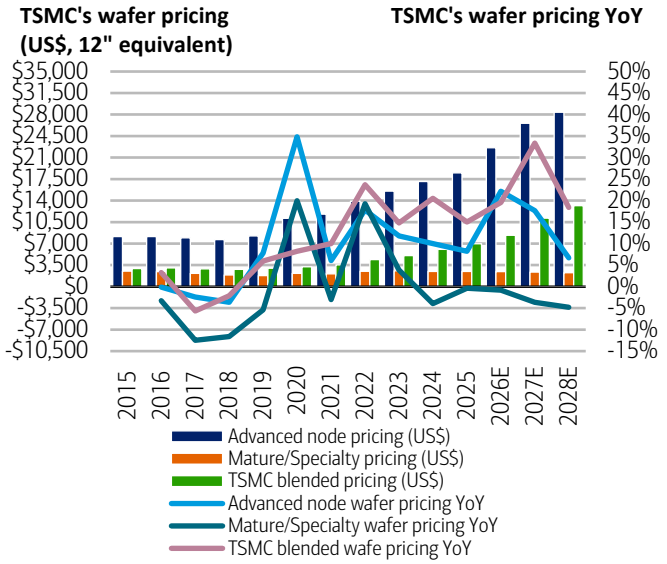
BofA GLOBAL RESEARCH

2026-27 sales and margins outlook solidified by healthy AI demand

With slightly better 4Q26 sales, we are lifting our US$ sales expectation for TSMC in '26 in line with the company's guidance of +44% YoY but keep 2027 estimates unchanged at +40% YoY. In addition to the existing drivers from AI accelerators (e.g. GPUs, ASICs, memory controllers), we see the demand strength is backed by a surge in demand for agentic AI CPUs as noted in our deep dive report supplemented by the upgrades in more high-end networking switches migrating to the most advanced nodes to enable faster data transmission in the complex AI data centers.

The strong cloud AI investment demand with expanding chipset categories is keeping the advanced node utilization tighter for longer. Along with inflationary cost pass through (e.g. equipment and consumables), TSMC should raise its advanced node and packaging pricing by 5-10% in the beginning of 2027 to reflect its value along with a mild 3-5% price hike on the mature nodes where the utilization is back to 90%+ as it optimizes the product mix following the strategic 6"/8" capacity retirement.

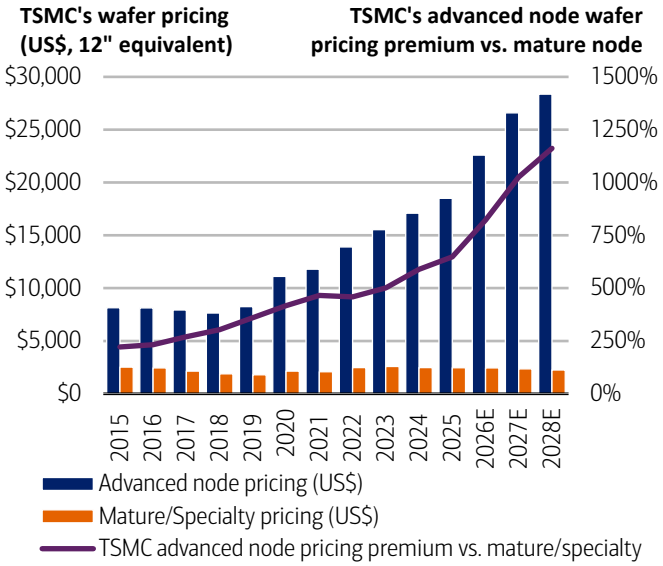
Exhibit 9: TSMC's wafer pricing by advanced vs. mature nodes
TSMC's wafer pricing by advanced vs. mature nodes



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 10: TSMC's wafer pricing gap for advanced vs. mature nodes
TSMC's advanced node pricing is now 9x higher than mature ones



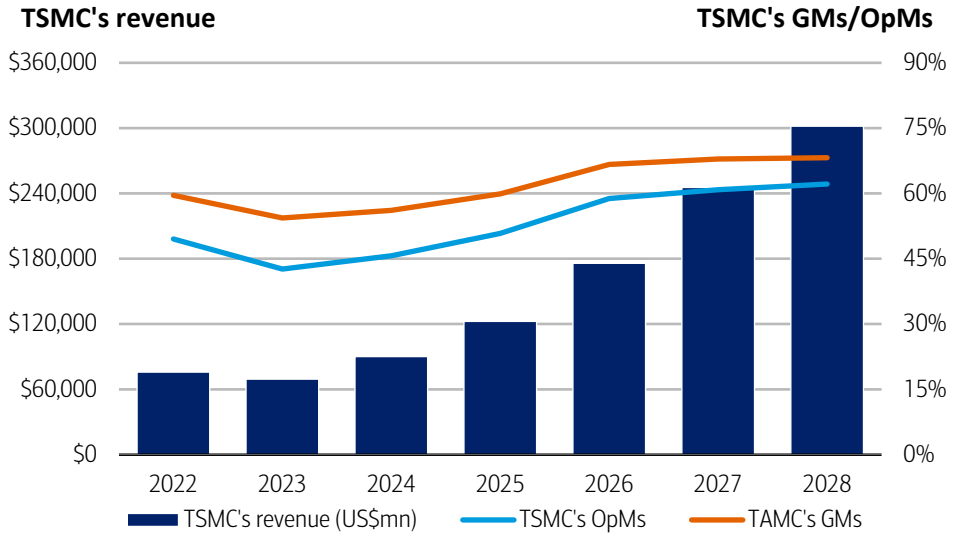
Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

However, factoring in the accelerated depreciation growth in 2027-28 (+24% CAGR from 2026-28) as more equipment is deployed in the new fabs and growing dilution from its overseas manufacturing footprint expansion and new node learning curve, we now model a more modest GMs step up to 67-68% in 2027-28 (vs. BofA prior 69-70%) following a jump to 66-67% in '26.

CR

Exhibit 11: TSMC's revenue and GMs/OpMs
We now model a more modest GMs step up to 67-68% in 2027-28 following a jump to 66-67% in '26



Source: BofA Global Research estimates, company data

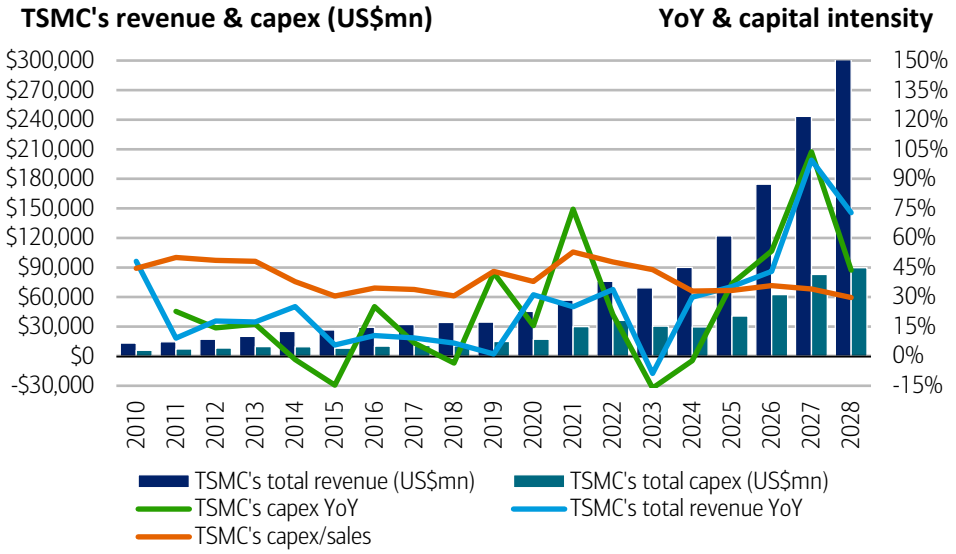
BofA GLOBAL RESEARCH

Capex on track for lifted guide for 2026 and will grow 20%+ CAGR from '26-28

TSMC on its 3Q26 earnings call sees the cloud AI chipset demand remains far above its current supply capability even incorporating the hyperscalers' investment, the financing environment, industry power availability and other component constraint, requiring the company to stay aggressive on capex in the next few years to grow capacity to try to narrow the supply/demand gap. While the company did not further raise capex for 2026, it reiterated the target of significantly higher spending from 2026-28 vs. the level in 2023-25. Despite tight equipment supply and price inflation in tools, materials and labor, the company emphasized the spending is driven by long term demand and it expects revenue (BofAe +31% CAGR from 2026-28) to outpace capex growth (BofAe +20% CAGR from 2026-28).

CR

Exhibit 12: TSMC's revenue and Capex trend
TSMC's sales should still outgrow Capex to keep its capital intensity manageable

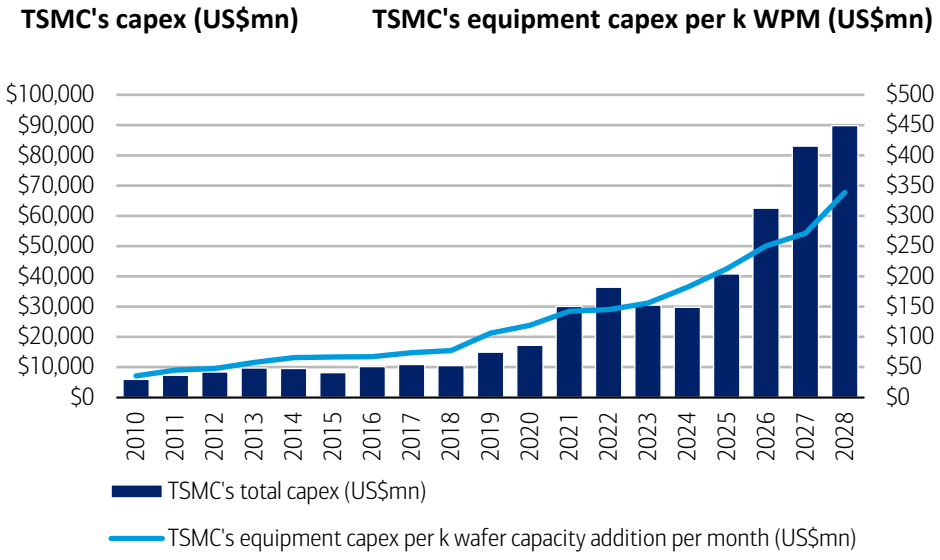


Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

Based on our bottom up capacity analysis across front-end manufacturing and back-end packaging, TSMC's capex spending is tracking to US$63bn for 2026 (vs. its latest guidance of US$60-64bn) mostly skewed toward advanced front-end manufacturing including 55-60k WPM 3nm capacity addition, 60k WPM 2nm capacity ramp, initial 3-5k WPM A16 capacity installation. The back-end packaging investment should be largely driven by its CoWoS capacity growth while the spending on the equipment pull-in for CPO and CoPoS remains limited as most of the equipment are still under sampling process for qualification and meaningful shipment will be in 2027-28.

Exhibit 13: TSMC's total Capex vs. equipment spending per k WPM capacity addition
TSMC spend 1-2x more on 1k WPM expansion due to its shift to advanced nodes in recent years



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

In addition to the earlier tool pull-in schedule due to tighter equipment supply with memory makers expanding, the fast-growing capex should reflect the rising cost per 1k WPM on the advanced nodes vs. mature nodes, with the equipment cost per 1k WPM advanced wafer capacity expansion has grown from US$156mn in 2023 to US$241mn in

CR

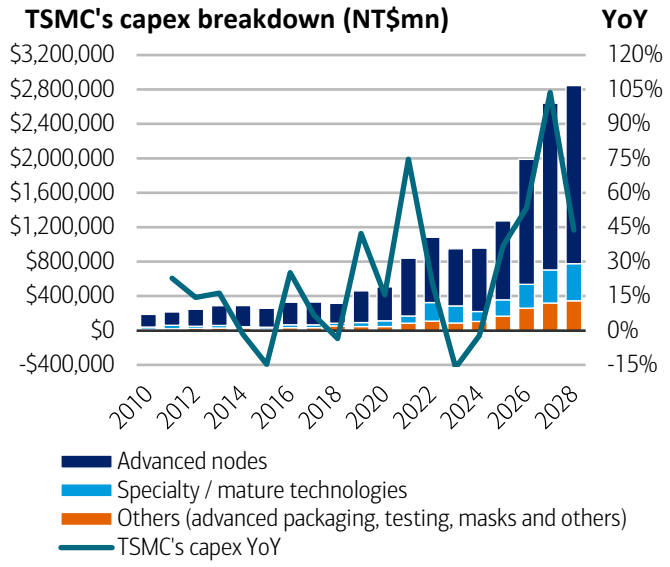
2025. Even factoring in more common use of equipment on the advanced nodes, the capex/1k WPM advanced capacity could further rise by 20% to US$289mn in 2027 with the technology architecture upgrade.

On top of the investment in the core front-end manufacturing, we expect the company to slightly outspend in the back-end packaging and estimate it is at the upper half of its guidance for 10-20% total capex, mainly for more aggressive capacity expansion to meet its customers' demand amid growing competition from Intel's EMIB-T solution and technology innovation. On the other hand, the maintenance capex could be reduced for specialty technologies as the company gradually retires the equipment.

Considering the capex required for further 3nm capacity expansion in the additional phases in its Tainan megafab, Arizona and the recently announced 2nd fab in Kumamoto on strong AI/HPC/automotive demand, ramping scale for 2nm family including A16, R&D on A14 supplemented by multiple back-end packaging technologies in progress (e.g. CoWoS, CoPoS, SoIC, WMCM) and lower capex efficiency in its US operation, our bottom up model shows TSMC's capex could be US$83bn/US$90bn in 2027/28 at a CAGR of 20% CAGR from 2026-28.

Beyond the technology upgrades and capacity expansion, we would note the equipment price hike would drive additional spending required by TSMC in the next few years. Our feedback from the tool suppliers noted they plan to hike the price by 10-15% for the front-end manufacturing equipment and 5-10% for the back-end packaging equipment effective from 2027 to reflect higher component and labor cost.

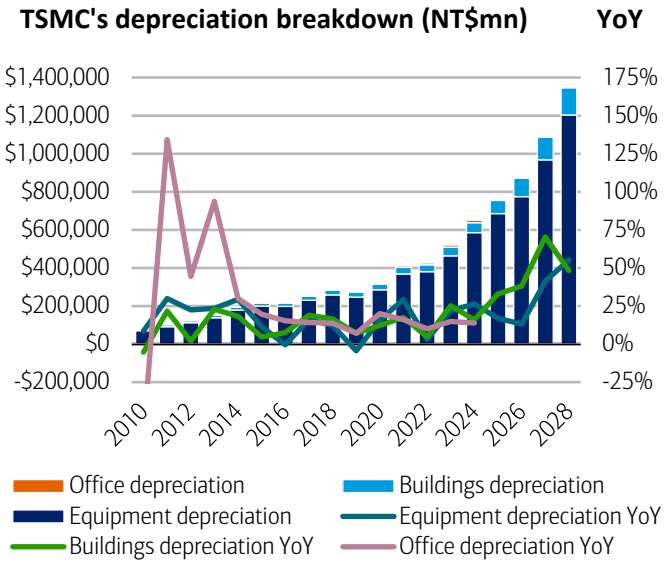
Exhibit 14: TSMC Capex spending breakdown by technology
TSMC spends the most on advanced front-end but packaging is outgrowing



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 15: TSMC's depreciation trend
TSMC's depreciation growth will be led by equipment



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

With TSMC's stronger bargaining power against the equipment supply chain vs. its peers, it should be able to fully pass on the ~10% higher nominal capex to its customers by lifting its foundry pricing when the tools are delivered for its production ramp in 2H27, setting up a good start for the pricing outlook in 2028.

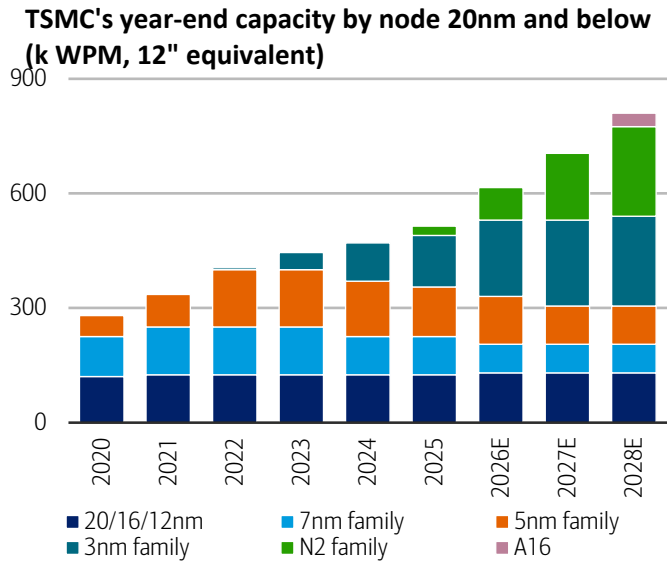
While the surge in headline capex hike could lead to concern on accelerating depreciation, we estimate it will only grow at 23% CAGR from 2026-28 due to modestly higher portion of the spending towards facility and clean room with much longer depreciation cycle than equipment. This should keep the depreciation still undergrowing its sales of +36% CAGR, sustaining GMs expansion to 68-69% levels by 2028.

CR

Advanced capacity expansion accelerates to support proliferating AI demand

As TSMC is in the process of retiring a portion of its equipment on 6" and 8" capacity while still adding the capacity on the advanced nodes, we estimate the company's total front-end manufacturing capacity will drop by 2% YoY in 2026.

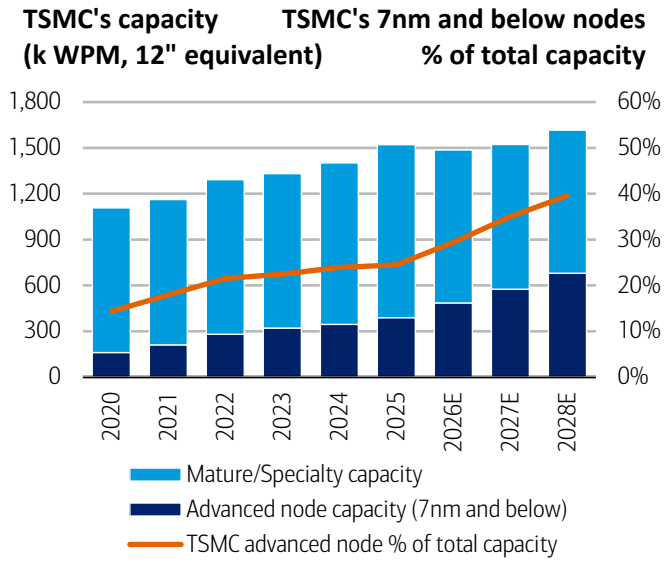
Exhibit 16: TSMC's year-end capacity by node for 20nm and below
TSMC will expand aggressively on 3nm and 2nm through 2027



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 17: TSMC's advanced node capacity as % of its installed base
TSMC's advanced nodes will reach 40% of its total capacity



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

The trend could be more stable into 2027-28 as TSMC strives to optimize its capacity mix and across node conversion to accommodate its customers' demand especially on the leading-edge technologies, with its advanced capacity set to grow by 10% CAGR from 2025-28 balanced by the mature capacity reduction at -8% CAGR.

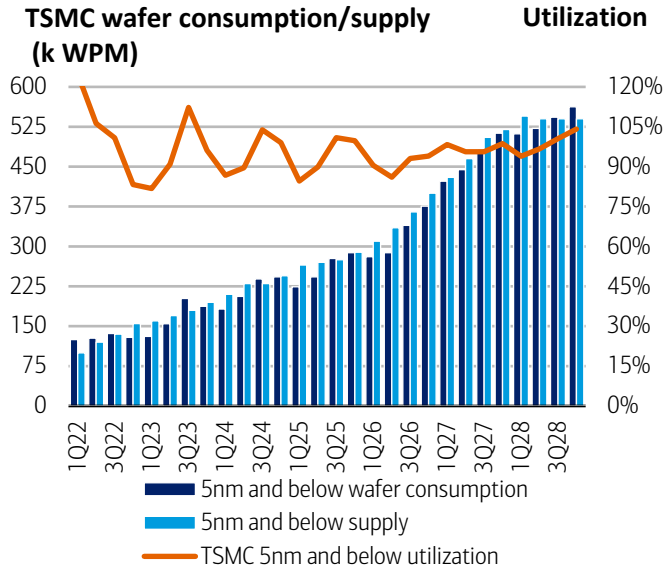
Based on our estimate, TSMC's capacity for 7nm and below nodes will grow from 320k WPM per month in 2023 to ~700k WPM by 2028 at a 16% CAGR, through expansion and efficient conversion from higher equipment commonality, cross technology planning and AI productivity gains (e.g. tool idle time and chamber cleaning schedule optimization). This will also support the mix of 7nm and below capacity climb toward 40-45% of its total capacity (vs. 24% in 2023). By region, we expect the capacity expansion through 2027 to enable its overseas fabs to account for 17% of TSMC's total capacity by 2028 (vs. 7% in 2020) and similarly for the advanced node capacity (vs. ~0% in 2023), mainly in Arizona and Kumamoto.

Specifically, we believe the company is on track to its target on 25% CAGR capacity expansion for its 5nm and 3nm from '22-27. We model its 3nm capacity will reach 190k in 4Q26 and grow to 230k WPM by 4Q27 through the expansion in its Tainan fab in 1H27 and Arizona 2nd phase in 3Q27. It is important to note TSMC's target to have another fab ready for 3nm production in Taian in 1H28 on top of its existing plan of deploying capacity in its new Japan fab would put the max capacity of the node up to 260k+ WPM at full scale, making it one of the largest nodes in TSMC's history.

In addition to N3 capacity scale up, on N2 family, we expect the installed base to grow by 70% CAGR from '26-28, with 1st year ramp delivering 45% more capacity vs. 3nm as it rolls out 5 fabs at once. Our estimate of 2nm capacity growing from 85k WPM in 4Q26 to 180k WPM by 4Q27, including 10k WPM A16 capacity, is consistent with TSMC's guidance of more than doubling its N2 capacity YoY in 2027.

CR

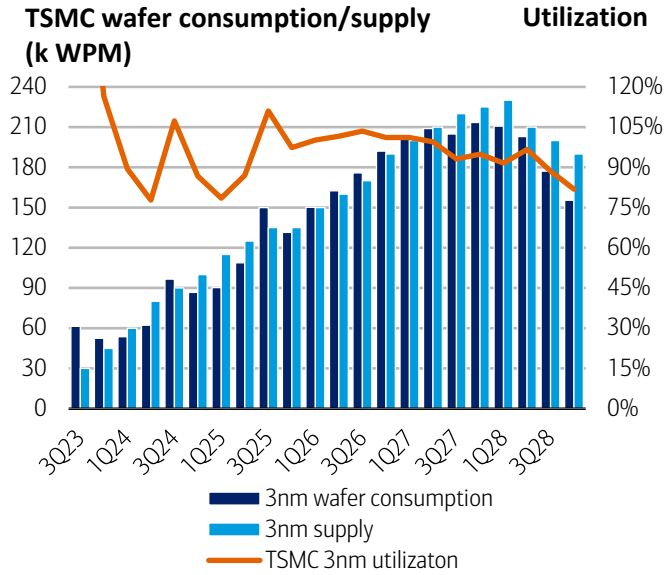
Exhibit 18: TSMC 5nm and below wafer consumption/supply/utilization
Utilization of 5nm and blow to remain high



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 19: TSMC 3nm wafer consumption/supply/utilization
TSMC 3nm to see utilization staying close to full



Source: Company Data, BofA Global Research estimates

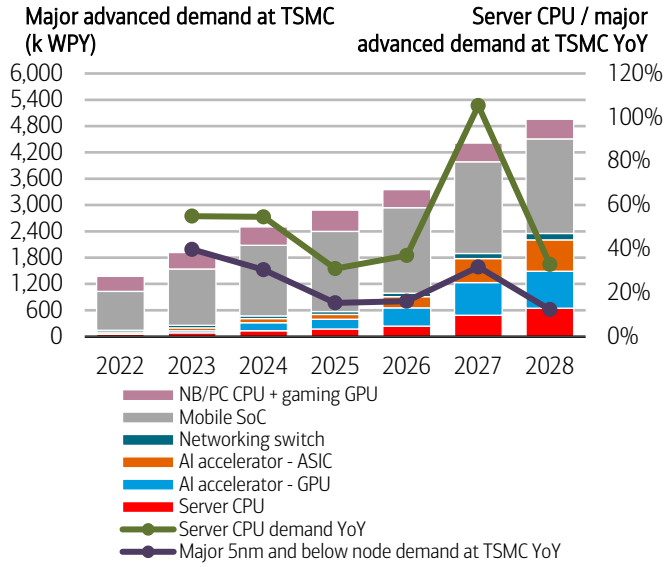
BofA GLOBAL RESEARCH

We believe TSMC is also on track for the technology and early stage production of its A14 node, the next major upgrade from N2, with installed base at 2-3k WPM now and set to reach pre-volume milestone in 2027 before entering into mass production in 2028.

On the demand, with 5nm and 3nm generating ~50% of TSMC's revenue in 2025, we analyze the supply and demand landscape through our bottom-up capacity model and wafer consumption by the company's major customers for their major products on these nodes. Based on our estimate, the utilization for 5nm and below nodes could stay high at least 90% if we only consider the effective capacity which is especially important during the period of frequent capacity conversion and new node ramp in the next few years.

CR

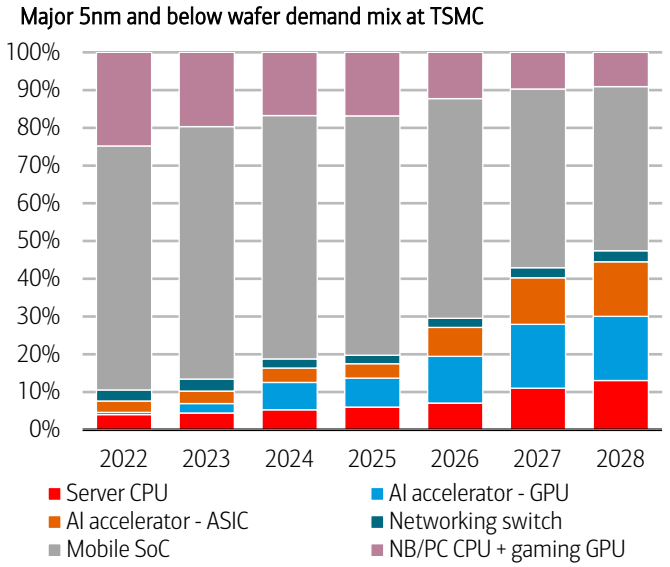
Exhibit 20: Major 5nm and below wafer demand at TSMC
Server CPU demand outgrows overall major advanced chipsets



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 21: Major 5nm and below wafer demand mix at TSMC
Server CPU accounts for 13% in 2028

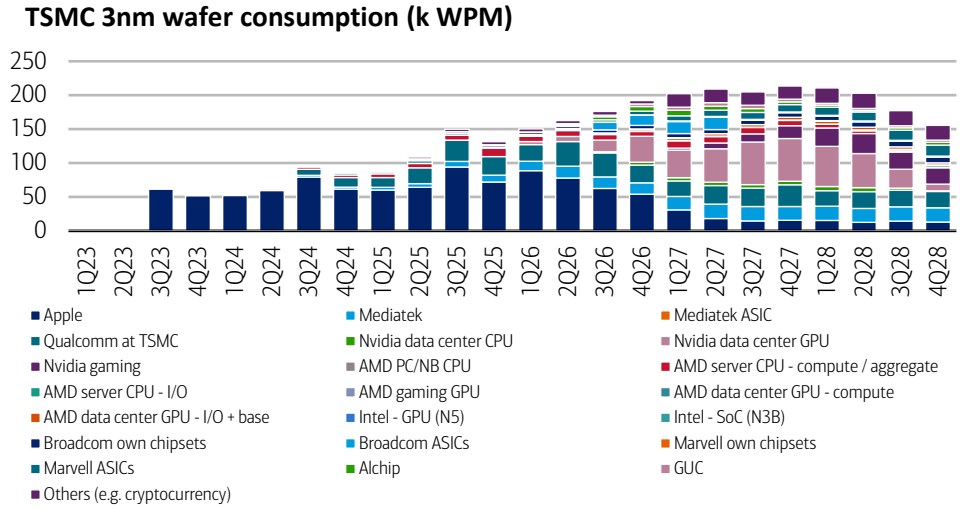


Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

We would note 3nm, as a sweet spot for high performance computing, networking and flagship smartphone chipsets, could see the utilization staying close to full even with continued expansion on strong project pipeline, sustaining continued price hike even in 2027 following the price surge in 2025-26. The node could be even more profitable from 2028-29 when the fixed cost starts rolling off following 5 year depreciation cycle.

Exhibit 22: TSMC 3nm wafer consumption by customers
Mediatek/Qualcomm/Nvidia should remain the largest customers following Apple's migration to 2nm



Source: BofA Global Research estimates

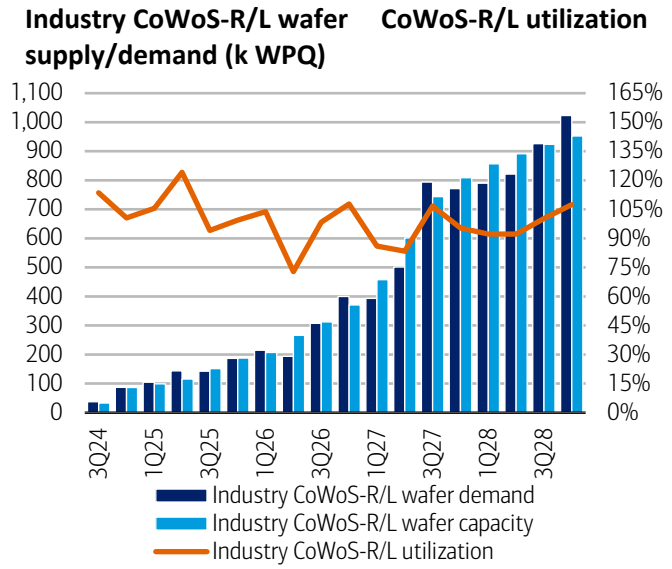
BofA GLOBAL RESEARCH

On the back-end packaging, with surging server CPU demand on top of the already strong order backlog for GPUs and ASICs, we estimate the industry's CoWoS supply will need to grow by 52% CAGR from 2026-28. TSMC will still lead the capacity expansion from 120k WPM in 4Q26 to 210k WPM in 4Q27, while the OSATs will also grow their portion of the supply with ASE and Amkor adding capacity to 70-75k+ WPM (vs. 30-35k in 4Q26) and 25-30k WPM (vs. 15-20k by 4Q26) in 4Q27 respectively. Specifically, while TSMC should still lead the supply for complex GPU and ASIC packaging, ASE grows as a major 2nd source with its capacity share growing from 18% of the industry in '26 to 29%

CR

in '28 and proven track record as the major CoWoS full process service provider for AMD's Venice server CPU.

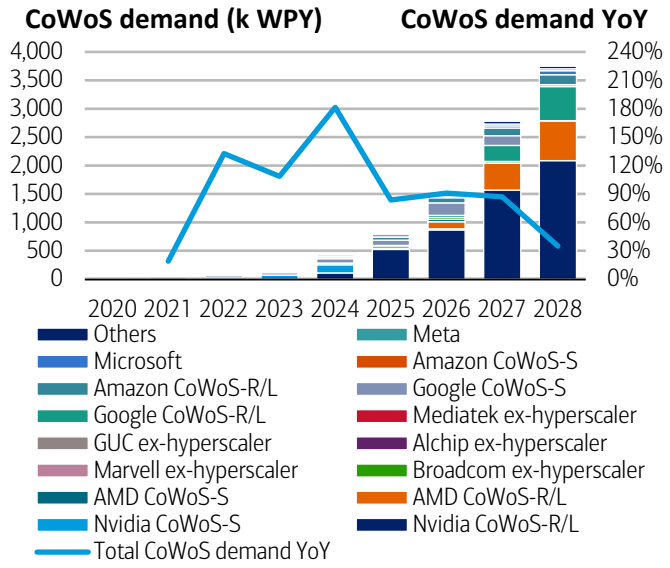
Exhibit 23: Industry CoWoS-R/L supply/demand
CoWoS-R/L utilization should stay at high levels



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 24: Industry CoWoS demand
CoWoS demand should remain strong through 2028



Source: Global Research estimates

BofA GLOBAL RESEARCH

We estimate TSMC's SoIC capacity will reach 20k WPM by 4Q26 and will need to grow to 35k WPM by 4Q27 and further to 50k WPM by 4Q28 with the applications proliferating from AMD's GPUs to Apple's M-series CPUs in 2026, hyperscalers' customized ARM based server CPUs in 2027 and adoption by Nvidia for its Feynman GPU along with the inflection of CPO penetration in 2028.

Similar to CoWoS, the SoIC technology could be even more important for TSMC going forward and the capacity ramp could be steep given much faster data transmission required for the HPC chipsets across CPUs, GPUs, AI accelerators and networking switches. At full utilization, we estimate SoIC could contribute US$2.9bn annually to TSMC in 2028 (1% of the company's total sales)

Discussion with Terafab ongoing, but operating control takes priority

TSMC on its earnings call confirmed a discussion with Terafab is ongoing as it also separately sent teams to assess the capacity expansion outside of Arizona, including Texas, with its US customers' request on more domestic capacity support. While there is no firm ownership structure as the evaluation is still early stage, TSMC emphasized the fabs will need to be operated by itself.

This is consistent with our view that, to have higher level of the control on capacity sufficiency, Terafab could consider to partner with TSMC but the likelihood of TSMC licensing technology or operating a fab with Terafab holding a meaningful share should be low. Specifically, we would note TSMC's JV fab record is limited only with WaferTech, SSMC, VIS, JASM, ESMC and AVSM mostly on mature nodes and it did not co-own a fab with Apple who contributed ~21% of its revenue from '15-25 mostly on the advanced processes as it keeps full business control and preserves customer neutrality.

Even if TSMC-Terafab discussion does not progress well, we believe the competition risk to TSMC is overstated as there could be a delay in the qualification process of Intel's 18A for external customers and 14A due to IP and manufacturing flow issues based on the recent supply chain feedback. In addition, the equipment vendors noted Terafab is only evaluating on equipment spec and potential procurement rather than placing firm

CR

orders. With extending equipment delivery lead time, this could put the company in an unfavorable position to receive tools for the initial production ramp in '29 it scheduled.

Exhibit 25: Wafer supply/demand analysis on Elon Musk's AI deployment target
With Terafab's near term delay and limited long term scale, TSMC will continue to be critical for Musk's ambitious AI compute deployment plan

Wafer supply/demand analysis on Elon Musk's AI deployment target	
SpaceX's annual orbital AI deployment target in 2028 (GW)	10
SpaceX's annual orbital AI deployment target in 2029 (GW)	100
Musk's long term target to deploy AI compute annual run rate (GW)	1,000
TDP of Nvidia's Feynman GPU (W)	6,000
GPU die output per front-end wafer (units)	55
Wafer demand for Feynman compute die for SpaceX's in '28 (k WPM)	10
Wafer demand for Feynman compute die for SpaceX's in '29 (k WPM)	101
Wafer demand for Feynman compute die to meet Musk's LT target (k WPM)	1,006
TSMC's 3nm and below capacity in 2028 (k WPM)	505
Terafab advanced wafer capacity scale by 2028 (k WPM)	0
Terafab capacity by 2036 based on its US$119bn investment plan (k WPM)	96

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

In the mid-term, Terafab's applications under Texas's JETI program earlier this year outlined ~US$119bn of planned investment across 4 phases from '26-36, with ~60% of it on tool procurement. The spending in '26-28 would be highly skewed toward facility construction before it starts to pull in US$6-8bn equipment per annum from '29-36.

We estimate the company will need to spend US$600mn+ for every 1k WPM expansion on the advanced nodes it targets in the US. Factoring the investment it targets to spend in the next decade, Terafab could only grow its capacity installed base by ~10k WPM per annum at maximum to 96k WPM front-end manufacturing capacity installed by '36, well below TSMC's scale, even if it can successfully deliver the technology roadmap through its own R&D effort and Intel partnership. This would prevent Terafab's operation from the economies of scale as the node migration continues, keeping TSMC as the major foundry partner for Tesla/SpaceX.

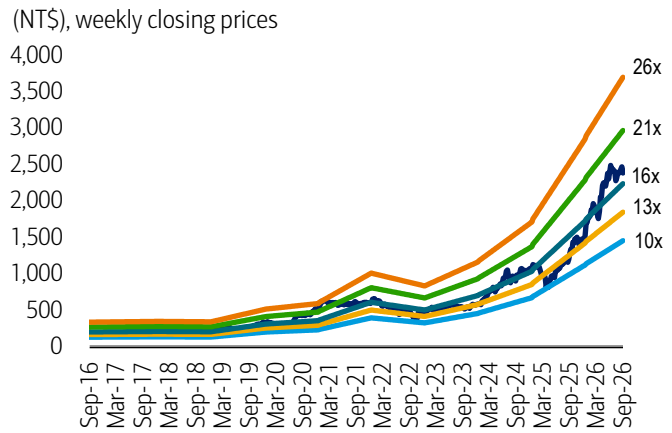
We also analyze the wafer consumption from SpaceX's annual orbital AI deployment target in '28/'29 of 10GW/100GW and Musk's long term 1TW target. Assuming the compute it plans to deploy in '28-29 is all powered by the most advanced Nvidia's Feynman GPUs delivering 6kW per package, this would drive 1.7mn/16.7mn chipset demand in '28/'29 respectively which requires 10k/101k monthly advanced wafer production.

From the supply perspective, as Feynman GPU will be built on TSMC's A16 node, the chipset demand should be fully accommodated. With limited support from Terafab, SpaceX in this scenario could become one of the largest customers for TSMC consuming ~20% of its advanced node capacity in '29 (vs. ~1% in '26 and ~2% in '28). Longer term, Musk's ambitious target to deploy 1 terawatt AI compute power annually would consume 1mn+ wafers per month on the advanced nodes.

CR

TSMC valuation reasonable and has room to re-rate with its HPC customers

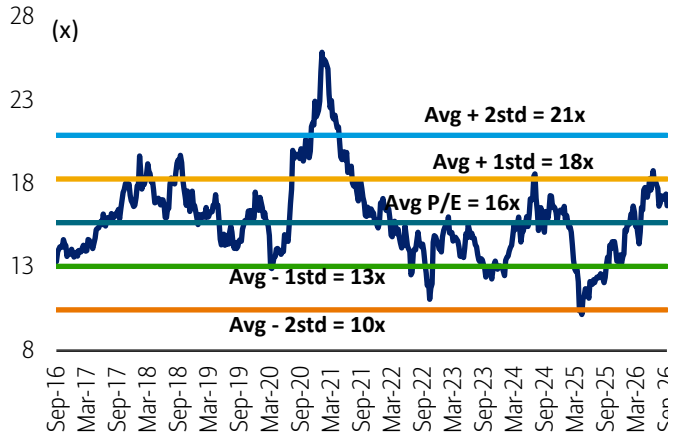
Exhibit 26: TSMC's P/E valuation band
TSMC's share at 15x '27E P/E is undemanding



Source: BofA Global Research estimates, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 27: TSMC's valuation based on BofA estimates
TSMC's valuation of 15x '27E P/E is at the middle of its historical 10-21x range

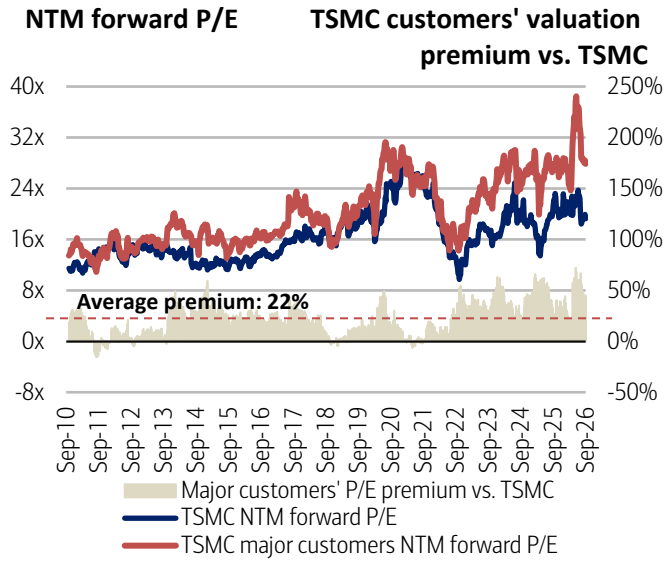


Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

TSMC's current valuation remains well supported when viewed through both its historical context and its relative positioning vs. its customers. In the past decade, TSMC's forward P/E has averaged ~16x with a range of 10-21x through the cycles.

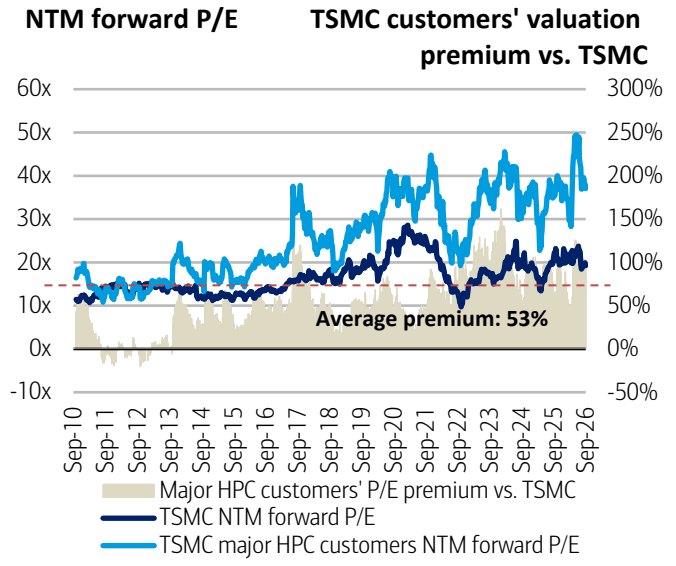
Exhibit 28: TSMC's P/E trend vs. its customers
TSMC customers' P/E is now at 45% premium to TSMC vs. 22% average



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 29: TSMC's P/E trend vs. its HPC customers
HPC fabless are the main driver for valuation premium at 100%



Source: BofA Global Research, Bloomberg

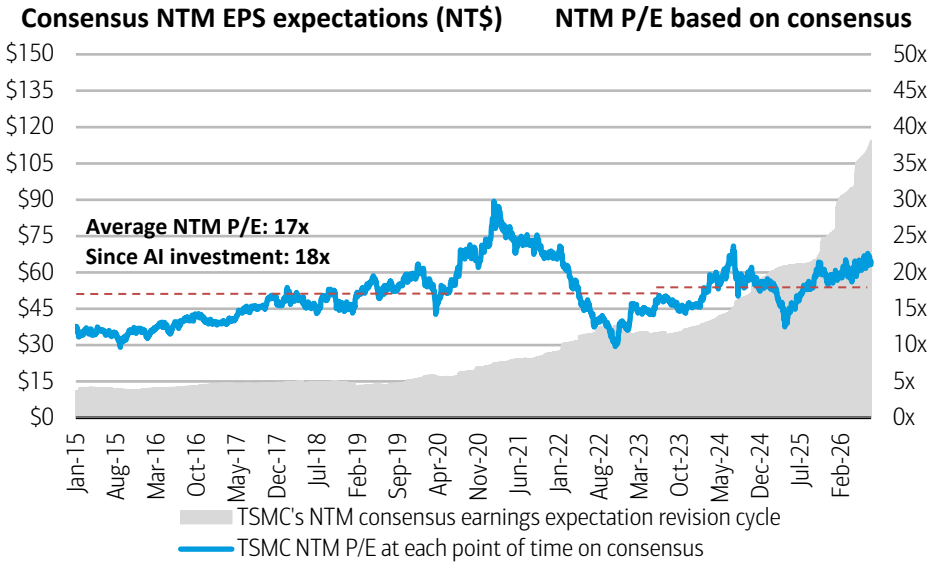
BofA GLOBAL RESEARCH

The relatively more modest valuation improvement for TSMC also contrasts with its customers' more pronounced re-ratings. Based on our analysis of forward P/E spreads, TSMC's major customers usually have a 22% average premium vs. TSMC in the long term though the current level is at 45%, suggesting TSMC's valuation has been lagging.

CR

Specifically, the valuation deviation for TSMC's HPC customers vs. the company has widened with 100-150% valuation premium vs. TSMC at this stage (vs. historical average of 53%). We believe this is driven by a surge in valuation for the direct beneficiaries from AI data center deployment, including the companies designing GPUs, ASICs and networking chipsets.

Exhibit 30: TSMC's consensus earnings vs. P/E trend
TSMC's share price strength only driven by earnings upgrade so far



Source: BofA Global Research, Bloomberg

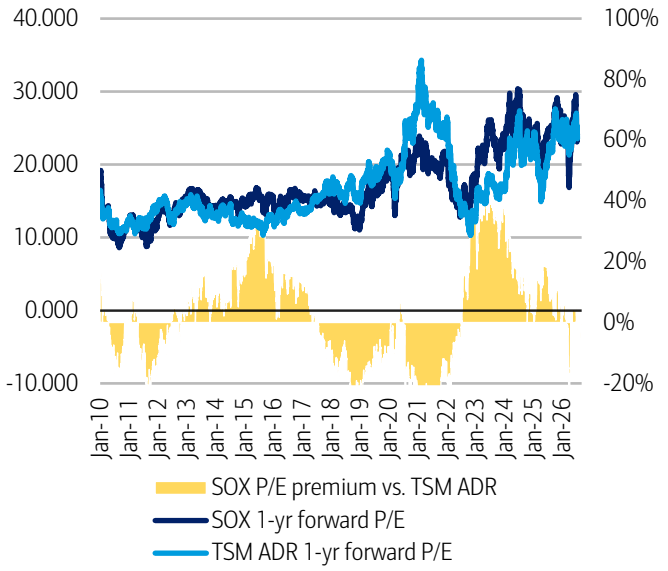
BofA GLOBAL RESEARCH

It is also noteworthy that, while TSMC's share performance since 2H23 has been solid, it is predominantly driven by the earnings upward revisions rather than multiple expansion. The company's NTM P/E has remained largely stable, only lifted to 18x in the past 2 years vs. its long-term ~17x levels. The company's ADR though has been performing better as it attracts a broader investor base vs. common share, with its premium still at 10% (vs. the long term average 8%) and at 20x 2027 consensus P/E.

With its firmer technology leadership, this should bode well for the company's earnings upgrade cycle and its share should also be re-rated at least 20x NTM P/E (24x P/E for ADR factoring the premium), back closer to its customers and the broader ecosystem.

CR

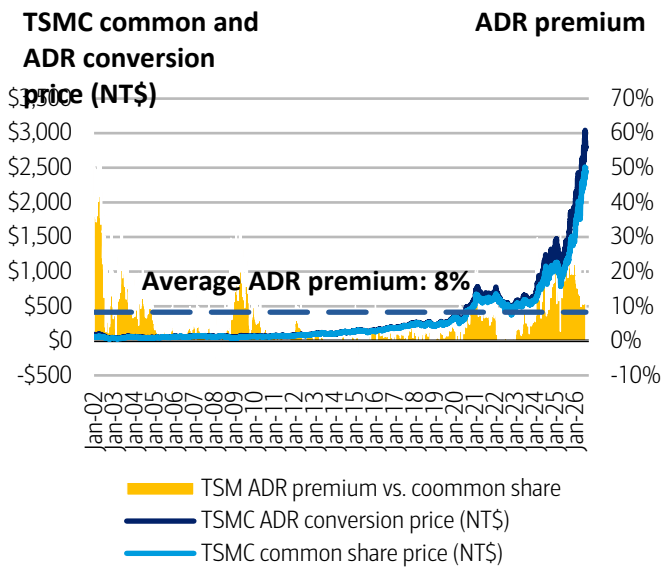
Exhibit 31: TSM ADR 1-year forward P/E vs. SOX premium
SOX index P/E is at -8% premium vs. TSM ADR



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 32: TSM ADR conversion price vs. common share and premium
TSM ADR premium is higher than common historical level at 10%



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 33: TSMC's quarterly and annual income statement
TSMC's quarterly and annual income statement

(NT$bn, %)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
Revenues	$1,134	$1,270	$1,494	$1,695	$1,711	$1,820	$2,068	$2,253	$3,809	$5,594	$7,853	$9,653
Cost of Sales	-$383	-$410	-$502	-$571	-$568	-$604	-$645	-$704	-$1,528	-$1,866	-$2,521	-$3,070
Gross Profit	$751	$860	$992	$1,124	$1,143	$1,217	$1,423	$1,549	$2,281	$3,728	$5,332	$6,583
Operating Expenses	-$92	-$94	-$116	-$135	-$134	-$136	-$141	-$143	-$345	-$437	-$554	-$583
Operating Income	$659	$767	$876	$989	$1,009	$1,080	$1,282	$1,406	$1,936	$3,291	$4,778	$6,000
Non-operating Income (Exp)	$29	$96	$35	$36	$39	$40	$45	$49	$106	$195	$173	$258
Pretax Income	$688	$862	$911	$1,025	$1,049	$1,121	$1,327	$1,455	$2,042	$3,486	$4,951	$6,258
Net tax	-$115	-$156	-$148	-$166	-$175	-$202	-$215	-$236	-$326	-$585	-$829	-$1,049
Net Income	$573	$707	$763	$859	$873	$918	$1,111	$1,219	$1,718	$2,901	$4,126	$5,212
Fully Diluted EPS - Local (NT$)	$22.08	$27.25	$29.43	$33.14	$33.70	$35.44	$42.89	$47.06	$66.24	$111.88	$159.09	$200.99
% of Revenue												
Gross Profit	66.2	67.7	66.4	66.3	66.8	66.8	68.8	68.8	59.9	66.6	67.9	68.2
Operating Expenses	-8.1	-7.4	-7.8	-8.0	-7.8	-7.5	-6.8	-6.3	-9.1	-7.8	-7.1	-6.0
Operating Income	58.1	60.3	58.7	58.3	59.0	59.3	62.0	62.4	50.8	58.8	60.8	62.2
Pretax Income	60.6	67.9	61.0	60.5	61.3	61.6	64.1	64.6	53.6	62.3	63.1	64.8
Net Income	50.5	55.6	51.1	50.7	51.0	50.5	53.7	54.1	45.1	51.9	52.5	54.0
QoQ Growth %												
Revenues	8.4	12.0	17.6	13.5	0.9	6.4	13.6	8.9				
Gross Profit	15.2	14.5	15.4	13.3	1.7	6.4	17.0	8.9				
Operating Income	16.7	16.3	14.3	12.8	2.1	7.0	18.7	9.7				
Net Income	13.3	23.4	8.0	12.5	1.7	5.2	21.0	9.7				
YoY Growth %												
Revenues	35.1	36.0	50.9	62.1	50.9	43.3	38.4	32.9	31.6	46.9	40.4	22.9
Gross Profit	52.3	57.2	68.6	72.4	52.2	41.4	43.4	37.8	40.4	63.4	43.0	23.5
Operating Income	61.9	65.4	75.0	75.1	53.2	40.9	46.3	42.2	46.4	70.0	45.2	25.6
Net Income	58.8	77.8	69.0	69.9	52.5	29.9	45.6	42.0	46.4	68.9	42.2	26.3
EPS	58.3	77.4	68.7	69.9	52.7	30.1	45.8	42.0	46.4	68.9	42.2	26.3

Source: Company data, BofA Global Research

BofA GLOBAL RESEARCH

CR

Exhibit 34: TSMC's quarterly and annual balance sheet
TSMC's quarterly and annual balance sheet

(NT$bn)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
Cash	$3,036	$3,134	$3,622	$4,045	$4,207	$4,535	$5,072	$5,875	$2,768	$4,045	$5,875	$8,786
Short Term Investment	$348	$384	$395	$389	$392	$391	$392	$391	$301	$389	$391	$391
Accounts receivable	$363	$441	$484	$344	$432	$443	$485	$434	$282	$344	$434	$482
Inventory	$311	$386	$428	$366	$437	$464	$484	$476	$288	$366	$476	$564
Other current assets	$207	$221	$157	$170	$187	$189	$185	$177	$179	$170	$177	$182
Current Assets	$4,266	$4,566	$5,085	$5,315	$5,655	$6,021	$6,617	$7,353	$3,817	$5,315	$7,353	$10,405
Long Term Investments	$167	$228	$197	$213	$205	$209	$207	$208	$172	$213	$208	$207
Fixed Assets	$3,955	$4,303	$4,526	$5,010	$5,412	$5,819	$6,207	$6,574	$3,692	$5,010	$6,574	$8,034
Other Assets	$274	$279	$244	$262	$265	$263	$259	$262	$252	$262	$262	$262
Long-term Assets	$4,395	$4,810	$4,968	$5,485	$5,882	$6,291	$6,672	$7,044	$4,116	$5,485	$7,044	$8,503
Total Assets	**$8,661**	**$9,376**	**#####**	**#####**	**$11,536**	**$12,312**	**$13,289**	**$14,397**	**$7,933**	**$10,800**	**$14,397**	**$18,909**
Short term loans	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Notes & Accounts Payable	$100	$111	$123	$105	$125	$135	$144	$142	$84	$105	$142	$179
Current portion of long term debt	$156	$167	$213	$168	$176	$181	$185	$178	$137	$168	$178	$180
Other current Liabilities	$1,458	$1,580	$1,632	$1,632	$1,648	$1,648	$1,648	$1,648	$1,237	$1,632	$1,648	$1,665
Current Liabilities	$1,714	$1,858	$1,968	$1,905	$1,950	$1,965	$1,977	$1,968	$1,458	$1,905	$1,968	$2,024
Long term debt	$901	$864	$820	$875	$867	$856	$854	$863	$896	$875	$863	$860
Other non-current liabilities	$113	$179	$222	$206	$180	$197	$201	$255	$118	$206	$255	$292
Long-term Liabilities	$1,014	$1,043	$1,042	$1,080	$1,047	$1,053	$1,056	$1,118	$1,014	$1,080	$1,118	$1,152
Total Liabilities	**$2,729**	**$2,901**	**$3,010**	**$2,986**	**$2,997**	**$3,018**	**$3,032**	**$3,086**	**$2,472**	**$2,986**	**$3,086**	**$3,176**
Total parent shareholders' equity	$5,891	$6,433	$6,998	$7,772	$8,497	$9,250	$10,214	$11,268	$5,420	$7,772	$11,268	$15,690
Minority Interest	$41	$42	$45	$42	$43	$43	$43	$43	$41	$42	$43	$43
Total Liabilities and Equity	**$8,661**	**$9,376**	**#####**	**#####**	**$11,536**	**$12,312**	**$13,289**	**$14,397**	**$7,933**	**$10,800**	**$14,397**	**$18,909**

Source: Company data, BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 35: TSMC's quarterly and annual cash flow
TSMC's quarterly and annual cash flow

(NT$bn)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
Net Profit	$572	$707	$763	$859	$874	$919	$1,112	$1,220	$1,718	$2,901	$4,126	$5,212
Depreciation & Amortization	$165	$199	$220	$248	$258	$269	$280	$314	$688	$832	$1,122	$1,272
Change in Working Capital	-$118	-$155	-$8	$170	-$155	-$30	-$50	$65	-$72	-$111	-$169	-$104
Other adjustment	$79	$33	-$55	$16	$2	-$6	$7	$25	-$59	$73	$28	$26
Operating Cash Flow	**$699**	**$783**	**$920**	**$1,293**	**$979**	**$1,153**	**$1,350**	**$1,625**	**$2,275**	**$3,695**	**$5,106**	**$6,406**
Capital Expenditure	-$351	-$496	-$443	-$698	-$664	-$664	-$664	-$664	-$1,272	-$1,988	-$2,657	-$2,875
Proceeds from sale of non-current assets	$2	$0	$0	$1	$1	$1	$1	$1	$1	$3	$3	$2
Acquisitions/Disposals of investments	-$30	-$37	$19	-$10	$5	-$2	$1	-$1	-$37	-$57	$3	$0
Other investment items	$22	$40	$157	-$18	-$3	$2	$4	-$4	$164	$201	$0	$0
Investing Cash Flow	**-$357**	**-$493**	**-$267**	**-$725**	**-$661**	**-$664**	**-$658**	**-$668**	**-$1,144**	**-$1,841**	**-$2,652**	**-$2,872**
Net Share issue/repurchase	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Dividends paid	-$130	-$156	-$156	-$156	-$156	-$156	-$156	-$156	-$467	-$596	-$622	-$622
Change in debt	$14	-$22	$1	$10	$0	-$6	$2	$2	$43	$4	-$2	-$1
Other financing cash flow	-$5	-$7	-$6	$0	$0	$0	$0	$0	-$17	-$18	$0	$0
Financing Cash Flow	**-$120**	**-$185**	**-$160**	**-$145**	**-$155**	**-$161**	**-$154**	**-$154**	**-$440**	**-$610**	**-$625**	**-$623**
Net Change in Cash	$222	$106	$493	$423	$162	$328	$537	$803	$690	$1,244	$1,830	$2,911
Exchange effect	$46	-$7	-$5	$0	$0	$0	$0	$0	-$50	$33	$0	$0
Cash, Beginning of Year	$2,768	$3,036	$3,134	$3,622	$4,045	$4,207	$4,535	$5,072	$2,128	$2,768	$4,045	$5,875
Cash, End of Year	$3,036	$3,134	$3,622	$4,045	$4,207	$4,535	$5,072	$5,875	$2,768	$4,045	$5,875	$8,786

Source: Company data, BofA Global Research

BofA GLOBAL RESEARCH

CR

Exhibit 36: Terms and definitions
Glossary of abbreviations/acronyms

Term	Definition
AI	Artificial Intelligence
AMD	Advanced Micro Devices
ASIC	Application Specific Integrated Circuit
AVSM	Advanced Vision Semiconductor Manufacturing
CoPoS	Chip-on-Panel-on-Substrate
CoWoS-L	Chip-on-Wafer-on-Substrate-Local Silicon Bridge
CoWoS-R	Chip-on-Wafer-on-Substrate-Redistribution Layer
CPO	Co-Packaged Optics
CPU	Central Processing Unit
EMIB	Embedded Multi-die Interconnect Bridge
ESMC	European Semiconductor Manufacturing Company
FoCoS	Fan-Out Chip on Substrate
GPU	Graphic Processing Unit
HBM	High Bandwidth Memory
HPC	High-Performance Computing
JASM	Japan Advanced Semiconductor Manufacturing
JV	Joint Venture
SoC	System on Chip
SoIC	System on Integrated Chips
SoW	System-on-Wafer
SPR	Super Power Rail
SSMC	Systems on Silicon Manufacturing Company
TPU	Tensor Processing Unit
VIS	Vanguard
WMCM	Wafer-level-multi-chip-module
WPM	Wafer per month

Source: BofA Global Research

BofA GLOBAL RESEARCH

Price objective basis & risk

Taiwan Semiconductor Manufacturing Co. (TSMWF / TSM)
Our PO for TSMC is NT$3,260 per share (US$620 per ADR), based on 20x 2027E P/E. TSMC's valuation has traded in the range of 10-26x PE from 2017. 20x is in the mid to high end of the historical range, above the average P/E of 16x. We think a PE above the average is justified given TSMC's enhanced industry position and mid-to-high 50% 5-year CAGR in AI should underpin a rerating

Downside risks to our price objective are: (1) greater-than-expected slowdown in global smartphone/consumer electronics demand, (2) Intel's potential insourcing strategy and ambitions in foundry service, (3) execution risks on advanced nodes, and (4) higher than expected tariffs.

Analyst Certification

I, Haas Liu, hereby certify that the views expressed in this research report accurately reflect my personal views about the subject securities and issuers. I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

Special Disclosures

Information on securities which are listed on the exchanges where ML Securities (Taiwan) Limited is not permitted to trade or solicit trades for clients is for informational purposes only and is not a recommendation or a solicitation to trade such securities. ML Securities (Taiwan) Limited will not execute transactions for nor accept orders from clients to trade such securities. Foreign investment in Taiwan securities is regulated and restricted. Currently, foreign investment in Taiwan securities is permitted by investment through: (1) global depository receipts, (2) convertible bonds, (3) mutual funds issued offshore of Taiwan, and (4) a special foreign institutional investors (FINIs) and foreign individual investors (FIDIs) program supervised by the Taiwan SFB whereunder FINIs/FIDIs may apply for investment ID to invest in Taiwan securities by registration with Taiwan Stock Exchange. FINIs will additionally need consent from the foreign exchange authority, ie, the Central Bank of China. In addition to the limitations above, various industry-specific percentage-based limitations on foreign ownership of Taiwan companies (and in some cases prohibitions) may apply. Investments are subject to exchange rate and currency conversion restrictions and risks. Dividends and interest earned by foreign investors' Taiwan securities/instruments are generally subject to a 20% withholding tax. Ordinary shares are not available to ML private client accounts in the U.S.

This report is distributed in Taiwan by Merrill Lynch Securities (Taiwan) Limited, which is regulated by the Taiwan SFB.

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Alchip	ALCPF	3661 TT	Haas Liu
	All Ring	XNQNF	6187 TT	Haas Liu
	AMEC	XRTOF	688012 CH	Dai Shen
	ASE Technology Holding	XSRIF	3711 TT	Haas Liu
	ASE Technology Holding -ADR	ASX	ASX US	Haas Liu
	ASMPT	ASMVF	522 HK	Simon Woo, CFA
	Aspeed	XLKMF	5274 TT	Mike Yang
	Chroma ATE	CRMJF	2360 TT	Haas Liu
	Doosan Corp	DOSAF	000150 KS	Simon Woo, CFA

CR

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	eMemory	XYLWF	3529 TT	Mike Yang
	GigaDevice	XGXIF	603986 CH	Daley Li, CFA
	Global Unichip Corp.	GBUHF	3443 TT	Haas Liu
	Grand Process Technology Corp	XZOWF	3131 TT	Haas Liu
	Hanmi Semiconductor	HNSIF	042700 KS	Simon Woo, CFA
	Hon Precision	XXPIF	7769 TT	Haas Liu
	Horizon Robotics	HRZRF	9660 HK	Daley Li, CFA
	Iluvatar CoreX	XSIBF	9903 HK	Daley Li, CFA
	Ingenic	XISCF	300223 CH	Dai Shen
	InnoScience Technology	XSCHF	2577 HK	Daley Li, CFA
	JCET Group Co Ltd	XJIEF	600584 CH	Dai Shen
	King Yuan Electronics Corp.	KYUFF	2449 TT	Haas Liu
	Kinik	KIKCF	1560 TT	Haas Liu
	Macronix International	MXICF	2337 TT	Dai Shen
	MediaTek	MDTKF	2454 TT	Haas Liu
	Montage Technology	XRDFF	688008 CH	Daley Li, CFA
	MPI Corporation	XMJCF	6223 TT	Mike Yang
	Nanya Technology	NNYAF	2408 TT	Simon Woo, CFA
	NCE Power	XVFFF	605111 CH	Daley Li, CFA
	OmniVision	XXHQF	603501 CH	Dai Shen
	Phison Electronics	PISNF	8299 TT	Simon Woo, CFA
	Powerchip Semiconductor Manufacturing Co	XCHPF	6770 TT	Haas Liu
	Powertech Technology	PWOGF	6239 TT	Simon Woo, CFA
	Rockchip	XRPXF	603893 CH	Daley Li, CFA
	Samsung C&T	XCDTF	028260 KS	Simon Woo, CFA
	Samsung Elec -G	SSNHZ	SMSN LI	Simon Woo, CFA
	Samsung Electronics	SSNLF	005930 KS	Simon Woo, CFA
	Samsung Electronics Preferred	SSNNF	005935 KS	Simon Woo, CFA
	Silergy Corp.	SLEGF	6415 TT	Mike Yang
	Silicon Motion	SIMO	SIMO US	Simon Woo, CFA
	SK Hynix	HXSCF	000660 KS	Simon Woo, CFA
	SK Hynix ADR	SKHY	SKHY US	Simon Woo, CFA
	SK Square	SKSQF	402340 KS	Simon Woo, CFA
	Taiwan Semiconductor Manufacturing Co.	TSM	TSM US	Haas Liu
	Taiwan Semiconductor Manufacturing Co.	TSMWF	2330 TT	Haas Liu
	Vanguard International Semiconductor Co	VGILF	5347 TT	Haas Liu
	Winbond Electronics	WBEKF	2344 TT	Dai Shen
	WinWay Technology	XWCLF	6515 TT	Mike Yang
	WT Microelectronics	XZOPF	3036 TT	Mike Yang
NEUTRAL				
	ASMedia Technology Inc.	XZSFF	5269 TT	Mike Yang
	Black Sesame Intl Holding	BSIHF	2533 HK	Daley Li, CFA
	Hangzhou Silan Microelectronics	XDFRF	600460 CH	Daley Li, CFA
	Lion Electronics	XDHFF	605358 CH	Dai Shen
	MetaX	XMEZF	688802 CH	Daley Li, CFA
	Novatek	NVKMF	3034 TT	Mike Yang
	Realtek	RLTKF	2379 TT	Haas Liu
	Shenzhen Goodix	XQPLF	603160 CH	Daley Li, CFA
	Soulbrain	XSBOF	357780 KS	Simon Woo, CFA
	VeriSilicon	XMLZF	688521 CH	Daley Li, CFA
UNDERPERFORM				
	Crystal Clear	XPPTF	300655 CH	Dai Shen
	Faraday	FDYTF	3035 TT	Haas Liu
	GlobalWafers	XWLFF	6488 TT	Mike Yang
	Hua Hong Grace	HHUSF	1347 HK	Dai Shen
	LX Semicon	XLXSF	108320 KS	Simon Woo, CFA
	M31 Technology	XMTZF	6643 TT	Mike Yang
	Maxscend	XMXSF	300782 CH	Daley Li, CFA
	Parade	PRDWF	4966 TT	Mike Yang
	United Microelectronics Corp.	XUMIF	2303 TT	Haas Liu
	United Microelectronics Corp.	UMC	UMC US	Haas Liu
	Win Semiconductors	WSMIF	3105 TT	Haas Liu
	Wonik IPS	XRHQF	240810 KS	Simon Woo, CFA

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst

iQmethod℠ Measures Definitions

Business Performance	Numerator	Denominator
Return On Capital Employed	NOPAT = (EBIT + Interest Income) × (1 – Tax Rate) + Goodwill Amortization	Total Assets – Current Liabilities + ST Debt + Accumulated Goodwill Amortization
Return On Equity	Net Income	Shareholders' Equity
Operating Margin	Operating Profit	Sales
Earnings Growth	Expected 5 Year CAGR From Latest Actual	N/A
Free Cash Flow	Cash Flow From Operations – Total Capex	N/A
Quality of Earnings	**Numerator**	**Denominator**
Cash Realization Ratio	Cash Flow From Operations	Net Income
Asset Replacement Ratio	Capex	Depreciation
Tax Rate	Tax Charge	Pre-Tax Income
Net Debt-To-Equity Ratio	Net Debt = Total Debt – Cash & Equivalents	Total Equity
Interest Cover	EBIT	Interest Expense
Valuation Toolkit	**Numerator**	**Denominator**
Price / Earnings Ratio	Current Share Price	Diluted Earnings Per Share (Basis As Specified)
Price / Book Value	Current Share Price	Shareholders' Equity / Current Basic Shares
Dividend Yield	Annualised Declared Cash Dividend	Current Share Price
Free Cash Flow Yield	Cash Flow From Operations – Total Capex	Market Cap = Current Share Price × Current Basic Shares
Enterprise Value / Sales	EV = Current Share Price × Current Shares + Minority Equity + Net Debt + Other LT Liabilities	Sales
EV / EBITDA	Enterprise Value	Basic EBIT + Depreciation + Amortization

iQmethod℠ is the set of BofA Global Research standard measures that serve to maintain global consistency under three broad headings: Business Performance, Quality of Earnings, and validations. The key features of iQmethod are: A consistently structured, detailed, and transparent methodology. Guidelines to maximize the effectiveness of the comparative valuation process, and to identify some common pitfalls.

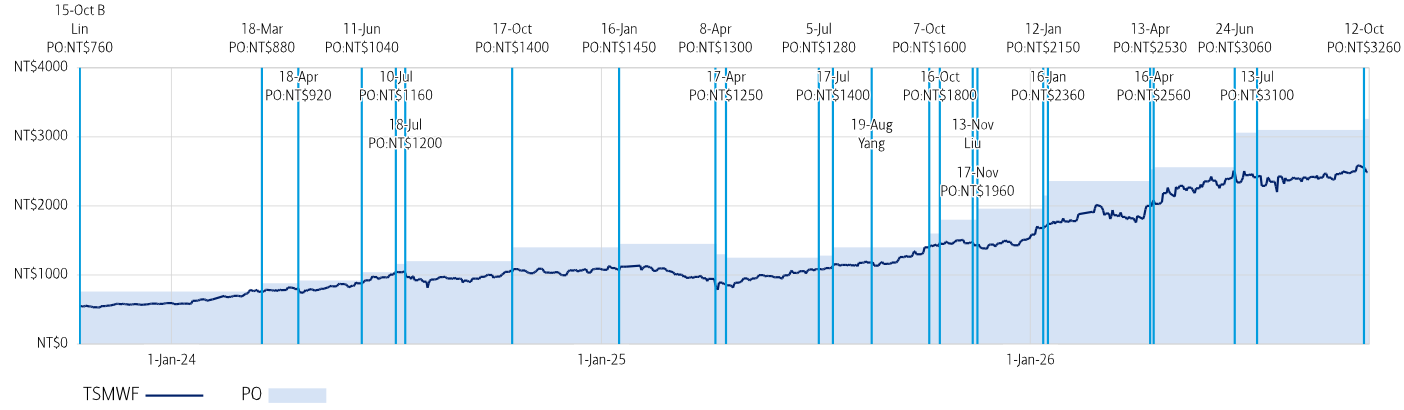
iQdatabase® is our real-time global research database that is sourced directly from our equity analysts' earnings models and includes forecasted as well as historical data for income statements, balance sheets, and cash flow statements for companies covered by BofA Global Research.

iQprofile℠, *iQmethod℠* are service marks of Bank of America Corporation. *iQdatabase®* is a registered service mark of Bank of America Corporation.

Disclosures

Important Disclosures

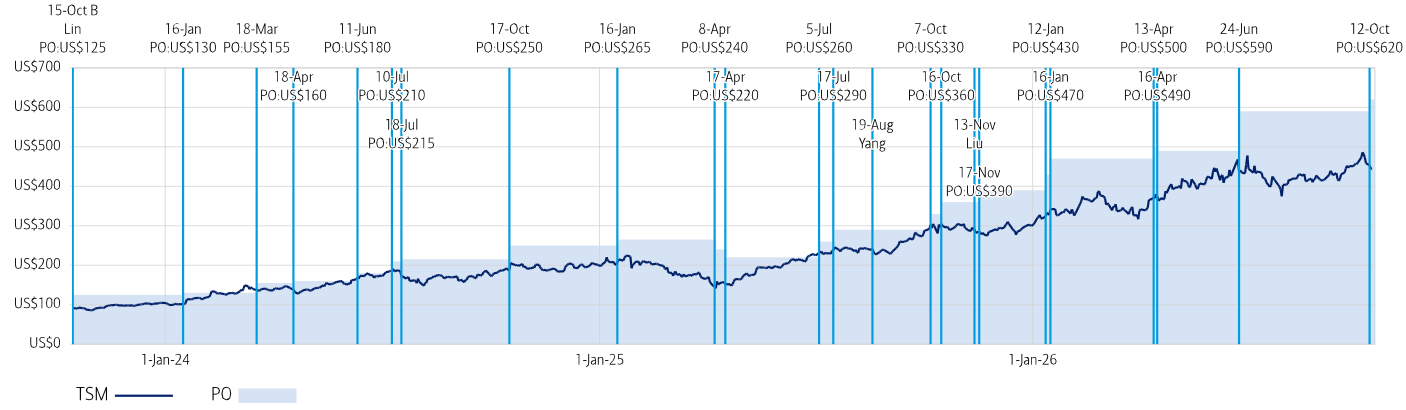
Taiwan Semi Mfg Co (TSMWF) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

Taiwan Semi Mfg Co (TSM) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

Equity Investment Rating Distribution: Technology Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	255	62.04%	Buy	157	61.57%
Hold	82	19.95%	Hold	46	56.10%
Sell	74	18.00%	Sell	31	41.89%

Equity Investment Rating Distribution: Global Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	2057	57.03%	Buy	1386	67.38%
Hold	806	22.35%	Hold	563	69.85%
Sell	744	20.63%	Sell	446	59.95%

[R1] Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

CR

FUNDAMENTAL EQUITY OPINION KEY: Opinions include a Volatility Risk Rating, an Investment Rating and an Income Rating. VOLATILITY RISK RATINGS, indicators of potential price fluctuation, are: A - Low, B - Medium and C - High. INVESTMENT RATINGS reflect the analyst's assessment of both a stock's absolute total return potential as well as its attractiveness for investment relative to other stocks within its Coverage Cluster (defined below). Our investment ratings are: 1 - Buy stocks are expected to have a total return of at least 10% and are the most attractive stocks in the coverage cluster; 2 - Neutral stocks are expected to remain flat or increase in value and are less attractive than Buy rated stocks and 3 - Underperform stocks are the least attractive stocks in a coverage cluster. An investment rating of 6 (No Rating) indicates that a stock is no longer trading on the basis of fundamentals. Analysts assign investment ratings considering, among other things, the 0-12 month total return expectation for a stock and the firm's guidelines for ratings dispersions (shown in the table below). The current price objective for a stock should be referenced to better understand the total return expectation at any given time. The price objective reflects the analyst's view of the potential price appreciation (depreciation).

Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster[R2]
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

[R2]Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

INCOME RATINGS, indicators of potential cash dividends, are: 7 - same/higher (dividend considered to be secure), 8 - same/lower (dividend not considered to be secure) and 9 - pays no cash dividend. *Coverage Cluster* is comprised of stocks covered by a single analyst or two or more analysts sharing a common industry, sector, region or other classification(s). A stock's coverage cluster is included in the most recent BofA Global Research report referencing the stock.

Price Charts for the securities referenced in this research report are available on the Price Charts website, or call 1-800-MERRILL to have them mailed.
BofAS or one of its affiliates acts as a market maker for the equity securities recommended in the report: Taiwan Semi Mfg Co.
The issuer is or was, within the last 12 months, an investment banking client of BofAS and/or one or more of its affiliates: Taiwan Semi Mfg Co.
BofAS or an affiliate has received compensation from the issuer for non-investment banking services or products within the past 12 months: Taiwan Semi Mfg Co.
The issuer is or was, within the last 12 months, a non-securities business client of BofAS and/or one or more of its affiliates: Taiwan Semi Mfg Co.
BofAS or an affiliate expects to receive or intends to seek compensation for investment banking services from this issuer or an affiliate of the issuer within the next three months: Taiwan Semi Mfg Co.
The country in which this issuer is organized has certain laws or regulations that limit or restrict ownership of the issuer's shares by nationals of other countries: Taiwan Semi Mfg Co.
BofAS or one of its affiliates is willing to sell to, or buy from, clients the common equity of the issuer on a principal basis: Taiwan Semi Mfg Co.
The issuer is or was, within the last 12 months, a securities business client (non-investment banking) of BofAS and/or one or more of its affiliates: Taiwan Semi Mfg Co.
BofA Global Research personnel (including the analyst(s) responsible for this report) receive compensation based upon, among other factors, the overall profitability of Bank of America Corporation, including profits derived from investment banking. The analyst(s) responsible for this report may also receive compensation based upon, among other factors, the overall profitability of the Bank's sales and trading businesses relating to the class of securities or financial instruments for which such analyst is responsible.

Other Important Disclosures

From time to time research analysts conduct site visits of covered issuers. BofA Global Research policies prohibit research analysts from accepting payment or reimbursement for travel expenses from the issuer for such visits.
Prices are indicative and for information purposes only. Except as otherwise stated in the report, for any recommendation in relation to an equity security, the price referenced is the publicly traded price of the security as of close of business on the day prior to the date of the report or, if the report is published during intraday trading, the price referenced is indicative of the traded price as of the date and time of the report and in relation to a debt security (including equity preferred and CDS), prices are indicative as of the date and time of the report and are from various sources including BofA Securities trading desks.
The date and time of completion of the production of any recommendation in this report shall be the date and time of dissemination of this report as recorded in the report timestamp.

Recipients who are not institutional investors or market professionals should seek the advice of their independent financial advisor before considering information in this report in connection with any investment decision, or for a necessary explanation of its contents.
Officers of BofAS or one or more of its affiliates (other than research analysts) may have a financial interest in securities of the issuer(s) or in related investments.
Refer to BofA Global Research policies relating to conflicts of interest.
"BofA Securities" includes BofA Securities, Inc. ("BofAS") and its affiliates. Investors should contact their BofA Securities representative or Merrill Global Wealth Management financial advisor if they have questions concerning this report or concerning the appropriateness of any investment idea described herein for such investor. "BofA Securities" is a global brand for BofA Global Research.
Information relating to Non-US affiliates of BofA Securities and Distribution of Affiliate Research Reports:
BofAS and/or Merrill Lynch, Pierce, Fenner & Smith Incorporated ("MLPF&S") may in the future distribute, information of the following non-US affiliates in the US (short name: legal name, regulator): Merrill Lynch (South Africa): Merrill Lynch South Africa (Pty) Ltd., regulated by the Financial Sector Conduct Authority; MLI (UK): Merrill Lynch International, regulated by the Financial Conduct Authority (FCA) and the Prudential Regulation Authority (PRA); BofASE (France): BofA Securities Europe SA is authorized by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers (AMF). BofA Securities Europe SA ("BofASE") with registered address at 51, rue La Boétie, 75008 Paris is registered under no 842 602 690 RCS Paris. In accordance with the provisions of French Code Monétaire et Financier (Monetary and Financial Code), BofASE is an établissement de crédit et d'investissement (credit and investment institution) that is authorised and supervised by the European Central Bank and the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers. BofASE's share capital can be found at www.bofaml.com/BofASEdisclaimer; BofA Europe (Milan): Bank of America Europe Designated Activity Company, Milan Branch, regulated by the Bank of Italy, the European Central Bank (ECB) and the Central Bank of Ireland (CBI); BofA Europe (Frankfurt): Bank of America Europe Designated Activity Company, Frankfurt Branch regulated by BaFin, the ECB and the CBI; BofA Europe (Zurich): Bank of America Europe Designated Activity Company, Zurich Branch, regulated by the Swiss Financial Market Supervisory Authority FINMA, the ECB and CBI; BofA Europe (Madrid): Bank of America Europe Designated Activity Company, Sucursal en España, regulated by the Bank of Spain, the ECB and the CBI; Merrill Lynch (Australia): Merrill Lynch Equities (Australia) Limited, regulated by the Australian Securities and Investments Commission; Merrill Lynch (Hong Kong): Merrill Lynch (Asia Pacific) Limited, regulated by the Hong Kong Securities and Futures Commission (HKSFC); Merrill Lynch (Singapore): Merrill Lynch (Singapore) Pte Ltd, regulated by the Monetary Authority of Singapore (MAS); Merrill Lynch (Canada): Merrill Lynch Canada Inc, regulated by the Canadian Investment Regulatory Organization; Merrill Lynch (Mexico): Merrill Lynch Mexico, SA de CV, Casa de Bolsa, regulated by the Comisión Nacional Bancaria y de Valores; BofAS Japan: BofA Securities Japan Co., Ltd., regulated by the Financial Services Agency; Merrill Lynch (Seoul): Merrill Lynch International, LLC Seoul Branch, regulated by the Financial Supervisory Service; Merrill Lynch (Taiwan): Merrill Lynch Securities (Taiwan) Ltd., regulated by the Securities and Futures Bureau; BofAS India: BofA Securities India Limited, regulated by the Securities and Exchange Board of India (SEBI); Merrill Lynch (Israel): Merrill Lynch Israel Limited, regulated by Israel Securities Authority; Merrill Lynch (DIFC): Merrill Lynch International (DIFC Branch), regulated by the Dubai Financial Services Authority (DFSA); Merrill Lynch (Brazil): Merrill Lynch S.A. Corretora de Títulos e Valores Mobiliários, regulated by Comissão de Valores Mobiliários; Merrill Lynch KSA Company: Merrill Lynch Kingdom of Saudi Arabia Company, regulated by the Capital Market Authority.
This information: has been approved for publication and is distributed in the United Kingdom (UK) to professional clients and eligible counterparties (as each is defined in the rules of the FCA and the PRA) by MLI (UK), which is authorized by the PRA and regulated by the FCA and the PRA - details about the extent of our regulation by the FCA and PRA are available from us on request; has been approved for publication and is distributed in the European Economic Area (EEA) by BofASE (France), which is authorized by the ACPR and regulated by the ACPR and the AMF; has been considered and distributed in Japan by BofAS Japan, a registered securities dealer under the Financial Instruments and Exchange Act in Japan, or its permitted affiliates; is issued and

CR

distributed in Hong Kong by Merrill Lynch (Hong Kong) which is regulated by HKSFC; is issued and distributed in Taiwan by Merrill Lynch (Taiwan); is issued and distributed in India by BofAS India; and is issued and distributed in Singapore to institutional investors and/or accredited investors (each as defined under the Financial Advisers Regulations) by Merrill Lynch (Singapore) (Company Registration No 198602883D). Merrill Lynch (Singapore) is regulated by MAS. Merrill Lynch Equities (Australia) Limited (ABN 65 006 276 795), AFS License 235132 (MLEA) distributes this information in Australia only to 'Wholesale' clients as defined by s.761G of the Corporations Act 2001. With the exception of Bank of America N.A., Australia Branch, neither MLEA nor any of its affiliates involved in preparing this information is an Authorised Deposit-Taking Institution under the Banking Act 1959 nor regulated by the Australian Prudential Regulation Authority. No approval is required for publication or distribution of this information in Brazil and its local distribution is by Merrill Lynch (Brazil) in accordance with applicable regulations. Merrill Lynch (DIFC) is authorized and regulated by the DFSA. Information prepared and issued by Merrill Lynch (DIFC) is done so in accordance with the requirements of the DFSA conduct of business rules. BofA Europe (Frankfurt) distributes this information in Germany and is regulated by BaFin, the ECB and the CBI. BofA Securities entities, including BofA Europe and BofASE (France), may outsource/delegate the marketing and/or provision of certain research services or aspects of research services to other branches or members of the BofA Securities group. You may be contacted by a different BofA Securities entity acting for and on behalf of your service provider where permitted by applicable law. This does not change your service provider. Please refer to the Electronic Communications Disclaimers for further information.

This information has been prepared and issued by BofAS and/or one or more of its non-US affiliates. The author(s) of this information may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so. BofAS and/or MLPF&S is the distributor of this information in the US and accepts full responsibility for information distributed to BofAS and/or MLPF&S clients in the US by its non-US affiliates. Any US person receiving this information and wishing to effect any transaction in any security discussed herein should do so through BofAS and/or MLPF&S and not such foreign affiliates. Hong Kong recipients of this information should contact Merrill Lynch (Asia Pacific) Limited in respect of any matters relating to dealing in securities or provision of specific advice on securities or any other matters arising from, or in connection with, this information. Singapore recipients of this information should contact Merrill Lynch (Singapore) Pte Ltd in respect of any matters arising from, or in connection with, this information. For clients that are not accredited investors, expert investors or institutional investors Merrill Lynch (Singapore) Pte Ltd accepts full responsibility for the contents of this information distributed to such clients in Singapore.

General Investment Related Disclosures:

Taiwan Readers: Neither the information nor any opinion expressed herein constitutes an offer or a solicitation of an offer to transact in any securities or other financial instrument. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or any other person without the express written consent of BofA Securities.

This document provides general information only, and has been prepared for, and is intended for general distribution to, BofA Securities clients. Neither the information nor any opinion expressed constitutes an offer or an invitation to make an offer, to buy or sell any securities or other financial instrument or any derivative related to such securities or instruments (e.g., options, futures, warrants, and contracts for differences). This document is not intended to provide personal investment advice and it does not take into account the specific investment objectives, financial situation and the particular needs of, and is not directed to, any specific person(s). This document and its content do not constitute, and should not be considered to constitute, investment advice for purposes of ERISA, the US tax code, the Investment Advisers Act or otherwise. Investors should seek financial advice regarding the appropriateness of investing in financial instruments and implementing investment strategies discussed or recommended in this document and should understand that statements regarding future prospects may not be realized. Any decision to purchase or subscribe for securities in any offering must be based solely on existing public information on such security or the information in the prospectus or other offering document issued in connection with such offering, and not on this document.

Securities and other financial instruments referred to herein, or recommended, offered or sold by BofA Securities, are not insured by the Federal Deposit Insurance Corporation and are not deposits or other obligations of any insured depository institution (including, Bank of America, N.A.). Investments in general and, derivatives, in particular, involve numerous risks, including, among others, market risk, counterparty default risk and liquidity risk. No security, financial instrument or derivative is suitable for all investors. Digital assets are extremely speculative, volatile and are not regulated in a manner similar to U.S. securities markets. In some cases, securities and other financial instruments may be difficult to value or sell and reliable information about the value or risks related to the security or financial instrument may be difficult to obtain. Investors should note that income from such securities and other financial instruments, if any, may fluctuate and that price or value of such securities and instruments may rise or fall and, in some cases, investors may lose their entire principal investment. Past performance is not necessarily a guide to future performance. Levels and basis for taxation may change.

This report may contain a short-term trading idea or recommendation, which highlights a specific near-term catalyst or event impacting the issuer or the market that is anticipated to have a short-term price impact on the equity securities of the issuer. Short-term trading ideas and recommendations are different from and do not affect a stock's fundamental equity rating, which reflects both a longer term total return expectation and attractiveness for investment relative to other stocks within its Coverage Cluster. Short-term trading ideas and recommendations may be more or less positive than a stock's fundamental equity rating.

BofA Securities is aware that the implementation of the ideas expressed in this report may depend upon an investor's ability to "short" securities or other financial instruments and that such action may be limited by regulations prohibiting or restricting "shortselling" in many jurisdictions. Investors are urged to seek advice regarding the applicability of such regulations prior to executing any short idea contained in this report.

Foreign currency rates of exchange may adversely affect the value, price or income of any security or financial instrument mentioned herein. Investors in such securities and instruments, including ADRs, effectively assume currency risk.

BofAS or one of its affiliates is a regular issuer of traded financial instruments linked to securities or digital assets that may have been recommended or discussed in this report. BofAS or one of its affiliates may, at any time, hold a trading position (long or short) in the securities and financial instruments recommended or discussed in this report.

BofA Securities, through business units other than BofA Global Research, may have issued and may in the future issue trading ideas or recommendations that are inconsistent with, and reach different conclusions from, the information presented herein. Such ideas or recommendations may reflect different time frames, assumptions, views and analytical methods of the persons who prepared them, and BofA Securities is under no obligation to ensure that such other trading ideas or recommendations are brought to the attention of any recipient of this information.

In the event that the recipient received this information pursuant to a contract between the recipient and BofAS for the provision of research services for a separate fee, and in connection therewith BofAS may be deemed to be acting as an investment adviser, such status relates, if at all, solely to the person with whom BofAS has contracted directly and does not extend beyond the delivery of this report (unless otherwise agreed specifically in writing by BofAS). If such recipient uses the services of BofAS in connection with the sale or purchase of a security referred to herein, BofAS may act as principal for its own account or as agent for another person. BofAS is and continues to act solely as a broker-dealer in connection with the execution of any transactions, including transactions in any securities referred to herein.

Copyright and General Information:

Copyright 2026 Bank of America Corporation. All rights reserved. iQdatabase® is a registered service mark of Bank of America Corporation. This information is prepared for the use of BofA Securities clients and may not be redistributed, retransmitted or disclosed, in whole or in part, or in any form or manner, without the express written consent of BofA Securities. This document and its content is provided solely for informational purposes and cannot be used for training or developing artificial intelligence (AI) models or as an input in any AI application (collectively, an AI tool). Any attempt to utilize this document or any of its content in connection with an AI tool without explicit written permission from BofA Global Research is strictly prohibited. BofA Global Research utilizes AI, including machine learning and other technologies, to enhance the services we provide to our clients. These technologies assist our analysts in various aspects of their work, including but not limited to data analysis, content extraction, content creation, data aggregation and summarization and identifying relevant information from diverse sources. All AI-driven processes are subject to review by BofA Global Research employees. BofA Global Research information is distributed simultaneously to internal and client websites and other portals by BofA Securities and is not publicly-available material. Any unauthorized use or disclosure is prohibited. Receipt and review of this information constitutes your agreement not to redistribute, retransmit, or disclose to others the contents, opinions, conclusion, or information contained herein (including any investment recommendations, estimates or price targets) without first obtaining express permission from an authorized officer of BofA Securities.

Materials prepared by BofA Global Research personnel are based on public information. Facts and views presented in this material have not been reviewed by, and may not reflect information known to, professionals in other business areas of BofA Securities, including investment banking personnel. BofA Securities has established information barriers between BofA Global Research and certain business groups. As a result, BofA Securities does not disclose certain client relationships with, or compensation received from, such issuers. To the extent this material discusses any legal proceeding or issues, it has not been prepared as nor is it intended to express any legal conclusion, opinion or advice. Investors should consult their own legal advisers as to issues of law relating to the subject matter of this material. BofA Global Research personnel's knowledge of legal proceedings in which any BofA Securities entity and/or its directors, officers and employees may be plaintiffs, defendants, co-defendants or co-plaintiffs with or involving issuers mentioned in this material is based on public information. Facts and views presented in this material that relate to any such proceedings have not been reviewed by, discussed with, and may not reflect information known to, professionals in other business areas of BofA Securities in connection with the legal proceedings or matters relevant to such proceedings.

This information has been prepared independently of any issuer of securities mentioned herein and not in connection with any proposed offering of securities or as agent of any issuer of any securities. None of BofAS any of its affiliates or their research analysts has any authority whatsoever to make any representation or warranty on behalf of the issuer(s). BofA Global Research

policy prohibits research personnel from disclosing a recommendation, investment rating, or investment thesis for review by an issuer prior to the publication of a research report containing such rating, recommendation or investment thesis.
Any information relating to sustainability in this material is limited as discussed herein and is not intended to provide a comprehensive view on any sustainability claim with respect to any issuer or security.
Any information relating to the tax status of financial instruments discussed herein is not intended to provide tax advice or to be used by anyone to provide tax advice. Investors are urged to seek tax advice based on their particular circumstances from an independent tax professional.
The information herein (other than disclosure information relating to BofA Securities and its affiliates) was obtained from various sources and we do not guarantee its accuracy. This information may contain links to third-party websites. BofA Securities is not responsible for the content of any third-party website or any linked content contained in a third-party website. Content contained on such third-party websites is not part of this information and is not incorporated by reference. The inclusion of a link does not imply any endorsement by or any affiliation with BofA Securities. Access to any third-party website is at your own risk, and you should always review the terms and privacy policies at third-party websites before submitting any personal information to them. BofA Securities is not responsible for such terms and privacy policies and expressly disclaims any liability for them.
All opinions, projections and estimates constitute the judgment of the author as of the date of publication and are subject to change without notice. Prices also are subject to change without notice. BofA Securities is under no obligation to update this information and BofA Securities ability to publish information on the subject issuer(s) in the future is subject to applicable quiet periods. You should therefore assume that BofA Securities will not update any fact, circumstance or opinion contained herein.
Subject to the quiet period applicable under laws of the various jurisdictions in which we distribute research reports and other legal and BofA Securities policy-related restrictions on the publication of research reports, fundamental equity reports are produced on a regular basis as necessary to keep the investment recommendation current.
Certain outstanding reports or investment opinions relating to securities, financial instruments and/or issuers may no longer be current. Always refer to the most recent research report relating to an issuer prior to making an investment decision.
In some cases, an issuer may be classified as Restricted or may be Under Review or Extended Review. In each case, investors should consider any investment opinion relating to such issuer (or its security and/or financial instruments) to be suspended or withdrawn and should not rely on the analyses and investment opinion(s) pertaining to such issuer (or its securities and/or financial instruments) nor should the analyses or opinion(s) be considered a solicitation of any kind. Sales persons and financial advisors affiliated with BofAS or any of its affiliates may not solicit purchases of securities or financial instruments that are Restricted or Under Review and may only solicit securities under Extended Review in accordance with firm policies.
Neither BofA Securities nor any officer or employee of BofA Securities accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this information.